Exploratory Data Analysis

Course: Computational Social Science, Homework #3
Team: FightClub
Research topic: Brainrot analysis
Lecturer: Andrew Kurochkin
Date: October 2026
Place: National University of Kyiv-Mohyla Academy, Faculty of Informatics

Time spent

Name	Time
Danylo	19h
Ivan	12h
Daryna	20h
Total	51h

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0. Overview

Since our goal is many-sided and has certain specific characteristics, we will follow our own flow with exploration and analysis of data. We already have hypotheses to test and we will try to find approaches in order to prove those hypothese right or wrong. We will not follow guidelines or try to check any boxes, instead we will follow our curiosity and play until we have the answer.

However, there is one guideline: before freestyling with analysis - explore the data and brainstorm what visualizations should be made.

Since we have 3 different hypotheses to test, the research is structured into 4 independent parts, each of them testing one hypothesis and doing separate data overview. Each hypothesis is tested with separate datasets and those datasets are reviewed independently in each of 4 parts. After that some of the results and findings are merged.

0.1 Hypotheses

Our topic is analysis of impact of doomscrolling and brainrot on people, discourse, their wellbeing and cognitive abilities over time. We study the effects of doomscrolling with the hypotheses below.

- H1: Discourse is getting shorter and simpler, accelerated by the short-form platforms
    - Lexical & NLP approach
    - Temporal/Behavioral aspect
- H2: Wellbeing is getting worse among doomscrollers
- H3: Crises intensify doomscrolling and negativity

0.2 Datasets

Each hypothesis is proven by using different datasets.

H1: Discourse is getting shorter and simpler, accelerated by the short-form platforms

Comments and submissions from reddit, collected from 01.2012 to 09.2026, divided into different categories.

1. Short-form: r/memes, r/teenagers
2. Long-form: r/books, r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

H2: Wellbeing is getting worse among doomscrollers

1. Social Media Addiction dataset
2. The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds
3. Sleep & Doomscrolling Habits Dataset
4. Social Media and Mental Health
5. Student Social Media & Mental Health
6. Reels/Shorts Consumption vs Attention Span

H3: Crises intensify doomscrolling and negativity

0.3 Additional approaches

- YouTube comments - work in progress

1 H1: Discourse is getting shorter and simpler, accelerated by the short-form platforms

In this section we test this hypothesis with data. We'll be analyzing the discourse of a dataset of 9.2 million comments and 1.8 million submissions that were posted on reddit between 2012 and 2026.

Before getting into data and into analysis, in order to pick the correct analysis path, let's take a bird's-eye view of our data and on our goal. We want to verify that over the 14 years discourse has been getting shorter and simpler. In order to check that, we have two approaches:

1. **Lexical & NLP approach**
2. **Temporal/Behavioral aspect**

In **Lexical & NLP approach** we analyse sentences and check the following metrics:

1. Vocabulary is shrinking
2. Sentences are getting shorter
    1. Words per comment
    2. Words per sentence
3. Lempel-Ziv algorithm
    1. Concatenation of comments into fixed-size chunks per subreddit per month

In **Temporal/Behavioral aspect** we analyze conversantions and look into:

1. Thread depth
    1. Thread size
    2. Share or replies vs top-level comments
    3. True depth

Ideas for analysis:

1. Readability (Flesch-Kincaid)
2. Avg word length

1.1. Overview

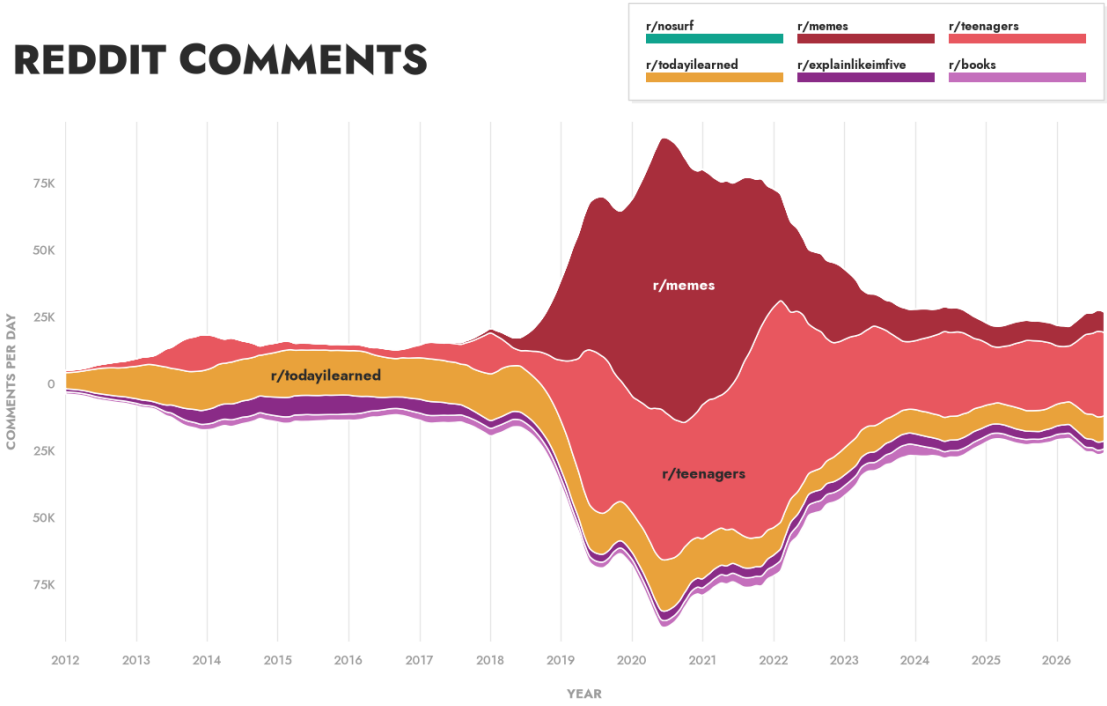
Our dataset consists of 9,237,205 comments and 1,870,257 posts, collected from 01.2012 to 09.2026, from these subreddits:

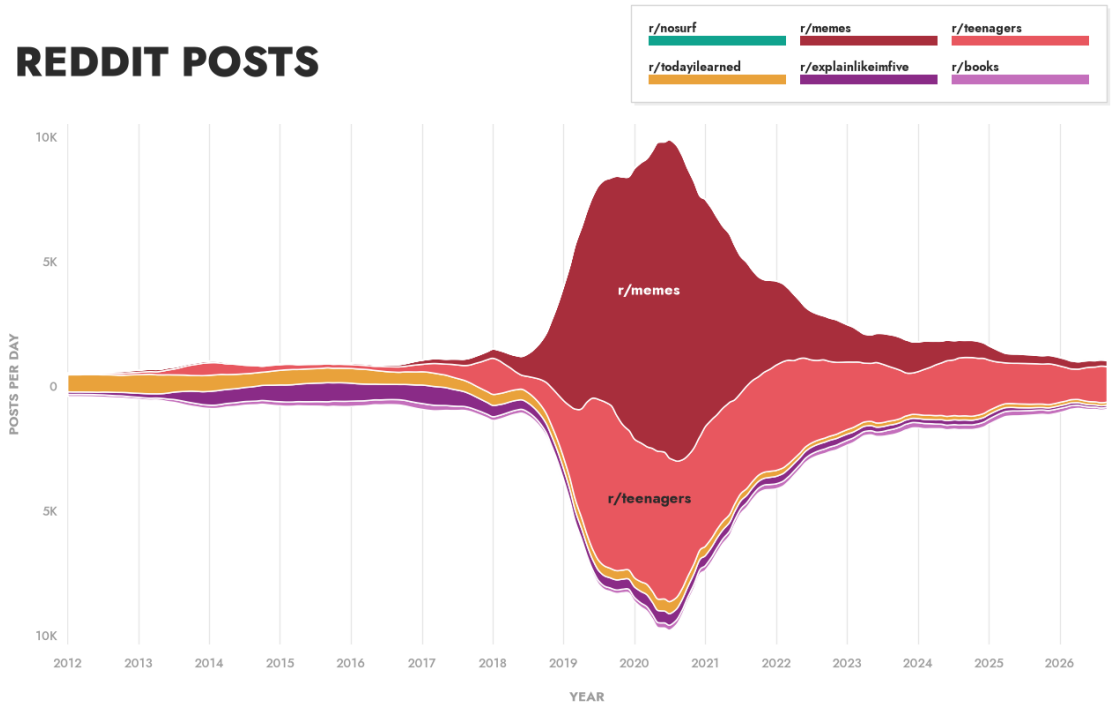
1. Short-form: r/memes, r/teenagers
2. Long-form: r/books, r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

Dataset was downloaded from Arctic Shift API. We didn't download full dataset of comments and posts because of the size and downloading time. Instead, we sampled 10,000 comments per subreddit per month and 2,000 posts per subreddit per month. Samples are downloaded from random time slots. Each month is cut into equal time slots, each slot holding roughly 1,000 comments or 200 posts. For our subreddits total amount of comments is 346.4 million and total amount of posts is 27.3 million. Our sample covers about 2.7% of all comments and 6.9% of all posts.

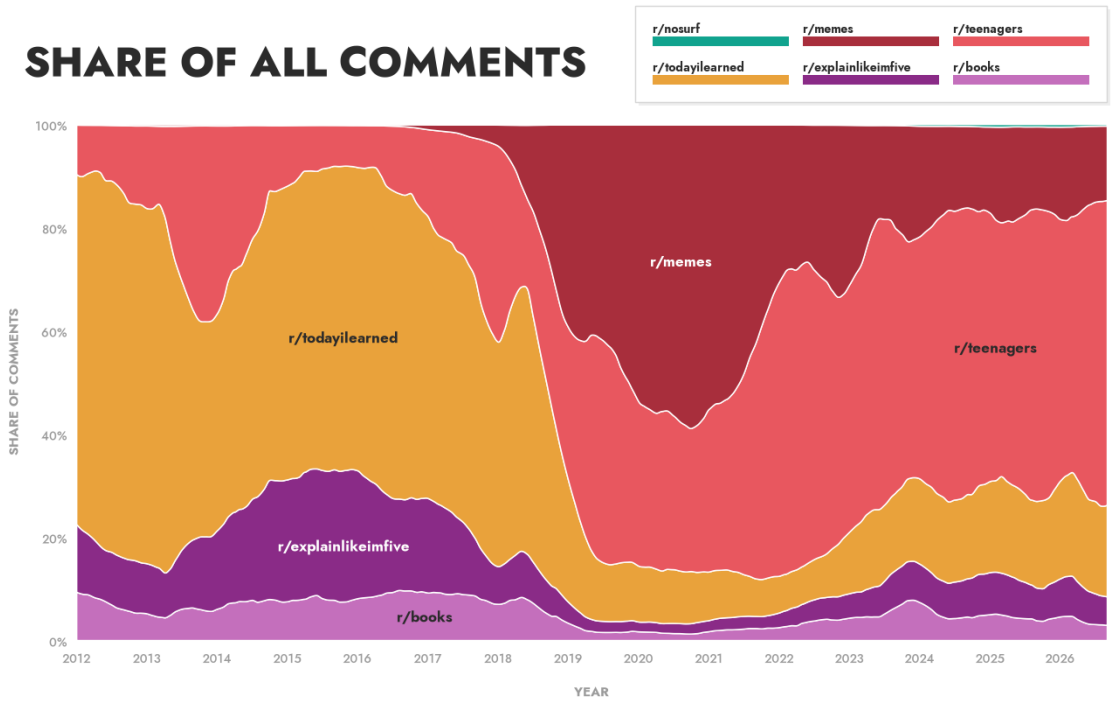
1.2 Activity and distribution

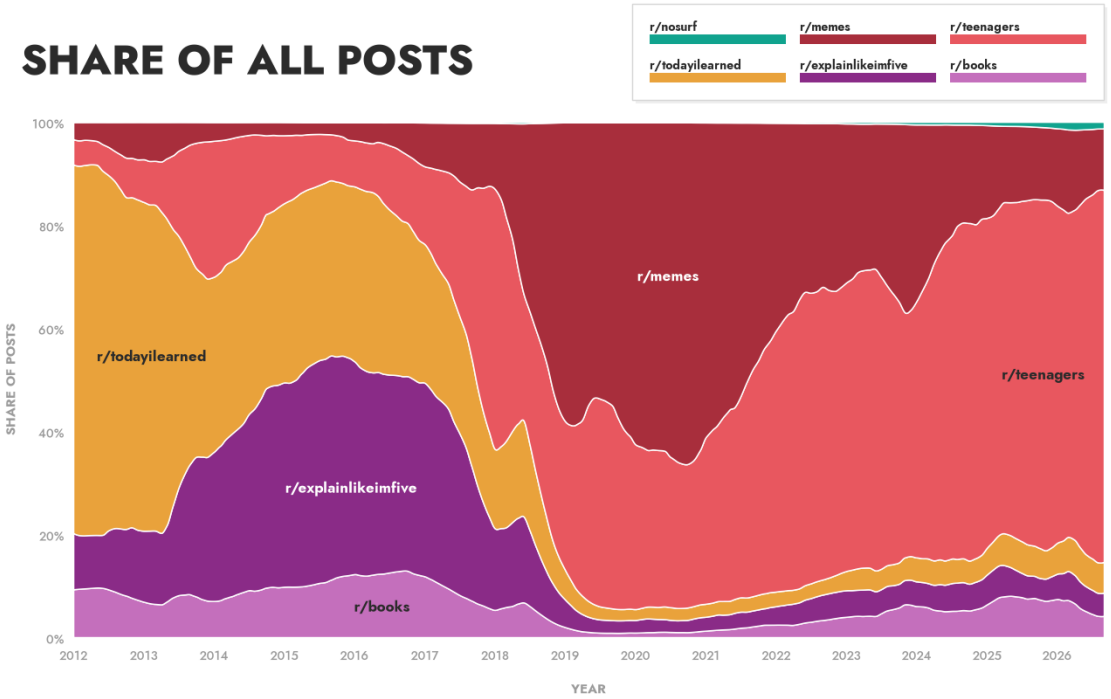
Before analysing the datasets, let's explore the activity over the last 14 years. First 2 plots represent the streamgraphs of comments and posts per day for all subreddit in absolute values. In August 2018 TikTok global, in March 2020 COVID-19 took place and TikTok went viral during the same period of time, as can seen on the streamgraph.



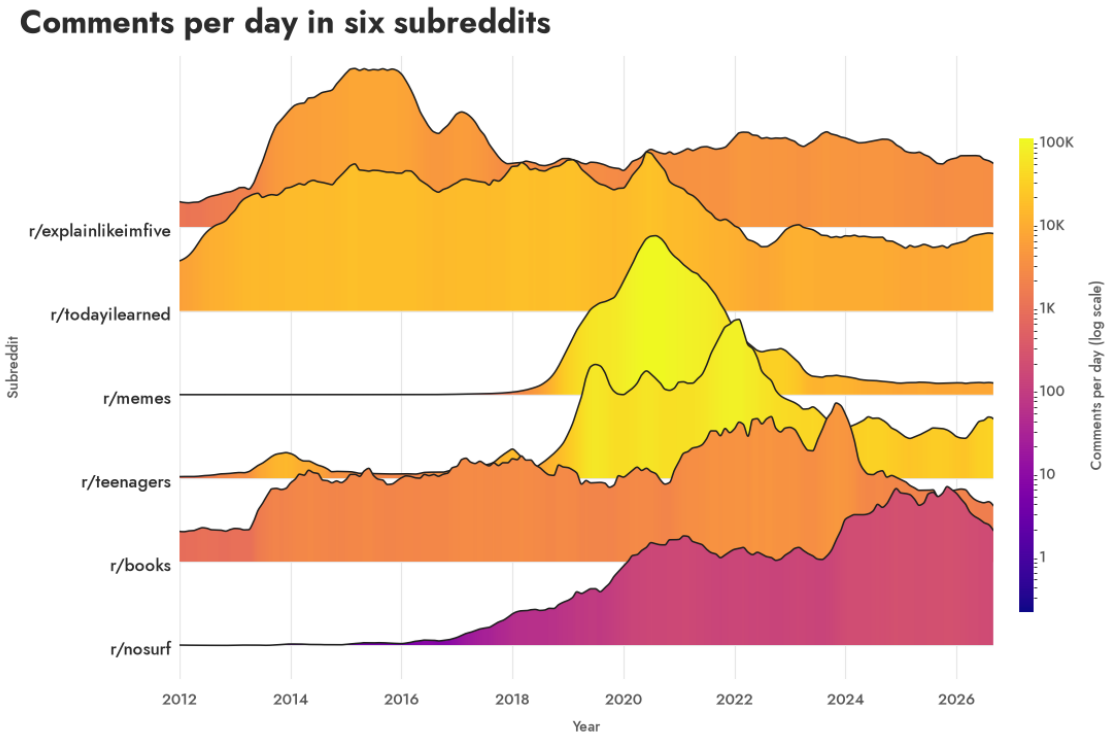


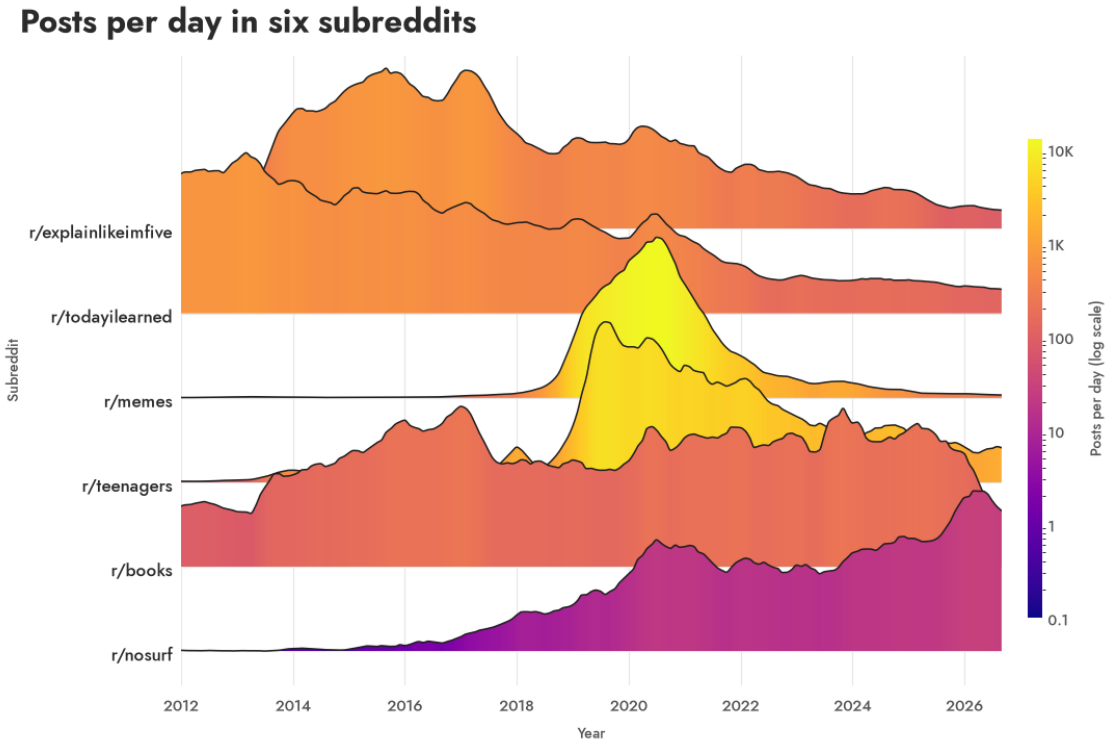
Let's take a look at stacked area chart which represents the activity of comments and posts per day for all subreddit, but in relative values.





Below is a ridgeline chart which represents the activity of each subreddit individually in gradient in absolute values.





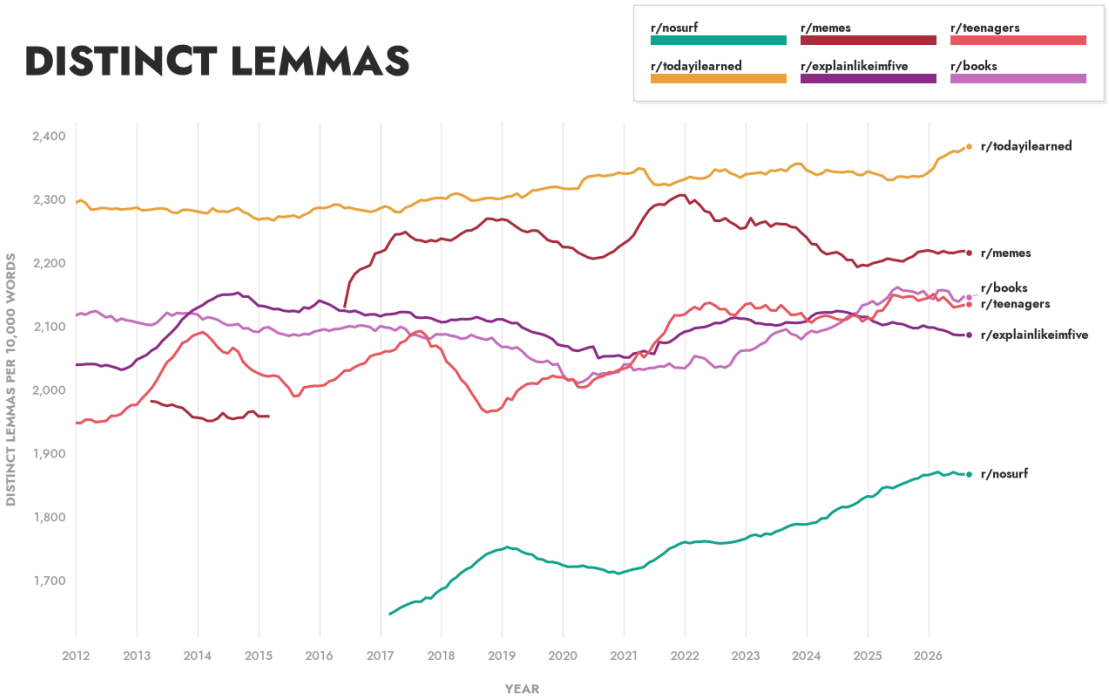
1.3 Lexical & NLP analysis

Let's analyze sentances with the following metrics:

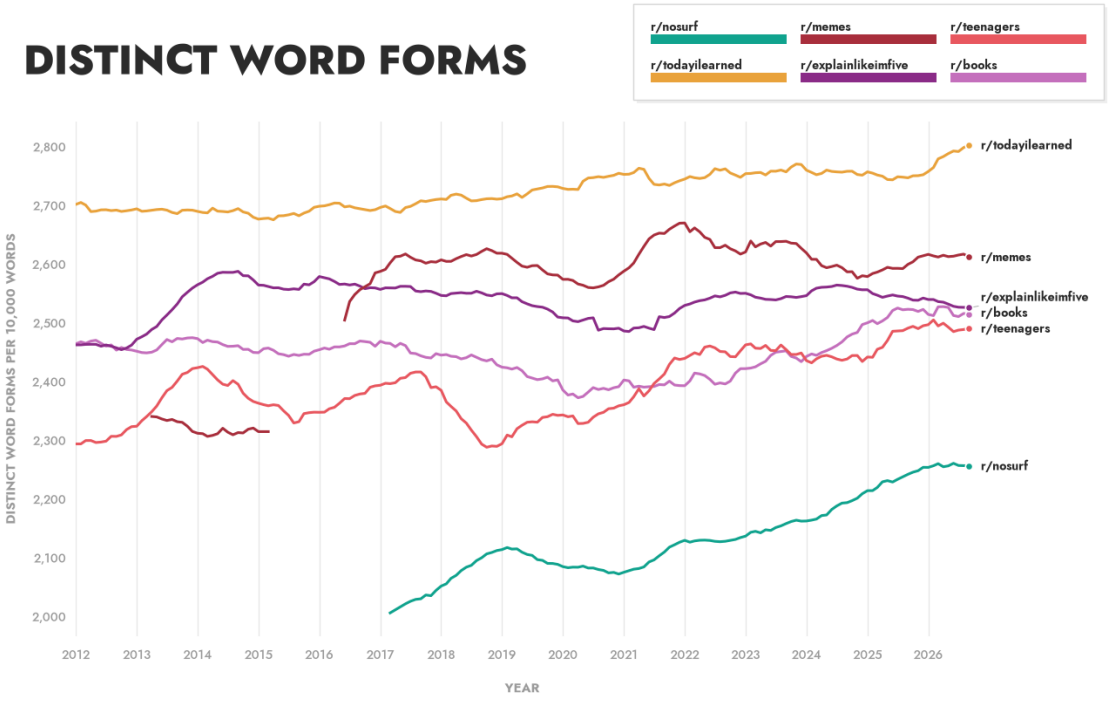
1. Vocabulary is shrinking
2. Sentences are getting shorter
    1. Words per comment
    2. Words per sentence
3. Lempel-Ziv algorithm
    1. Concatenation of comments into fixed-size chunks per subreddit per month

1.3.1. Shrinking vocabulary

Question is if people use fewer different words over time. In order to find that we measure distinct lemmas per 10,000 words per month per subreddit.

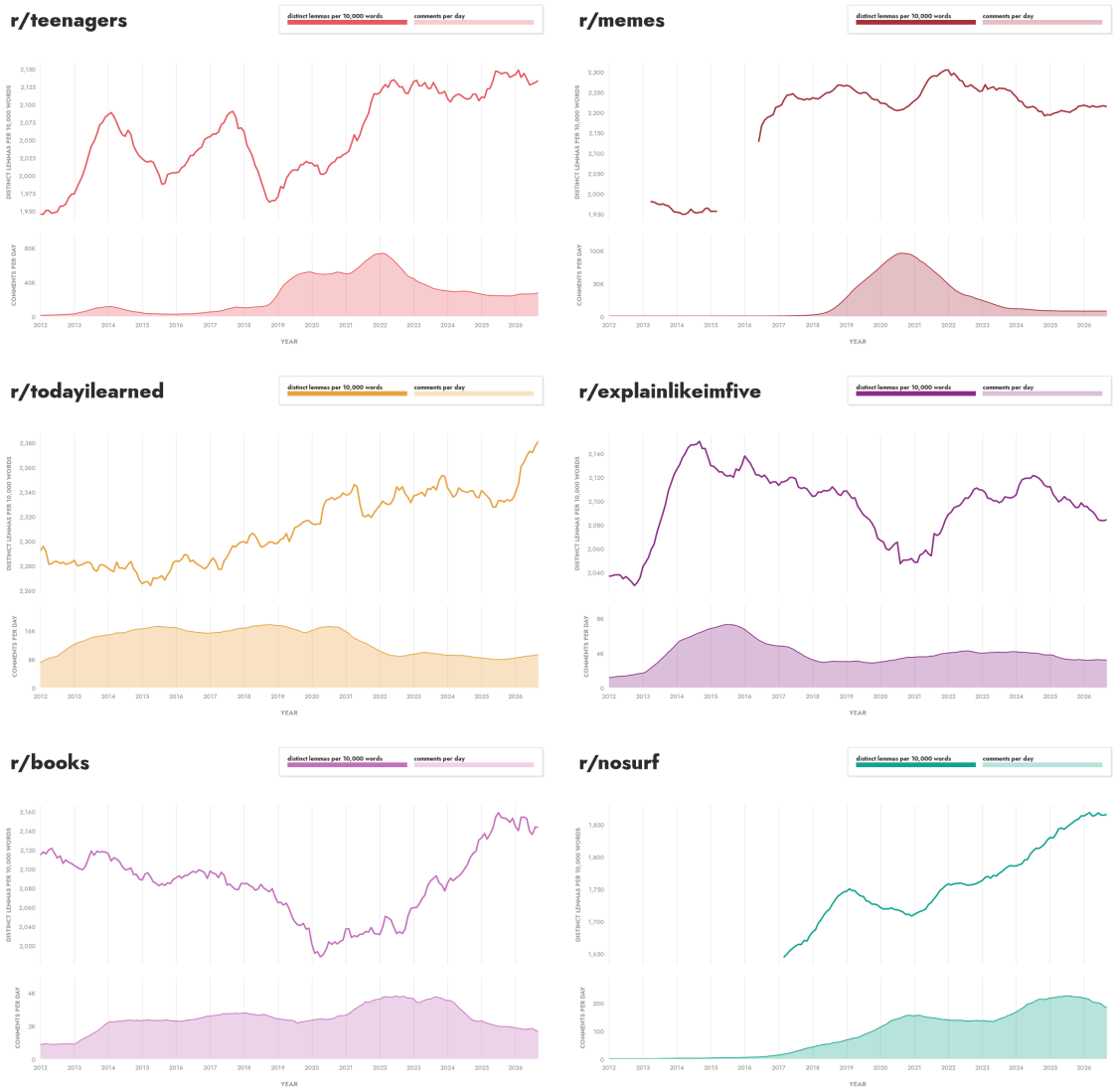


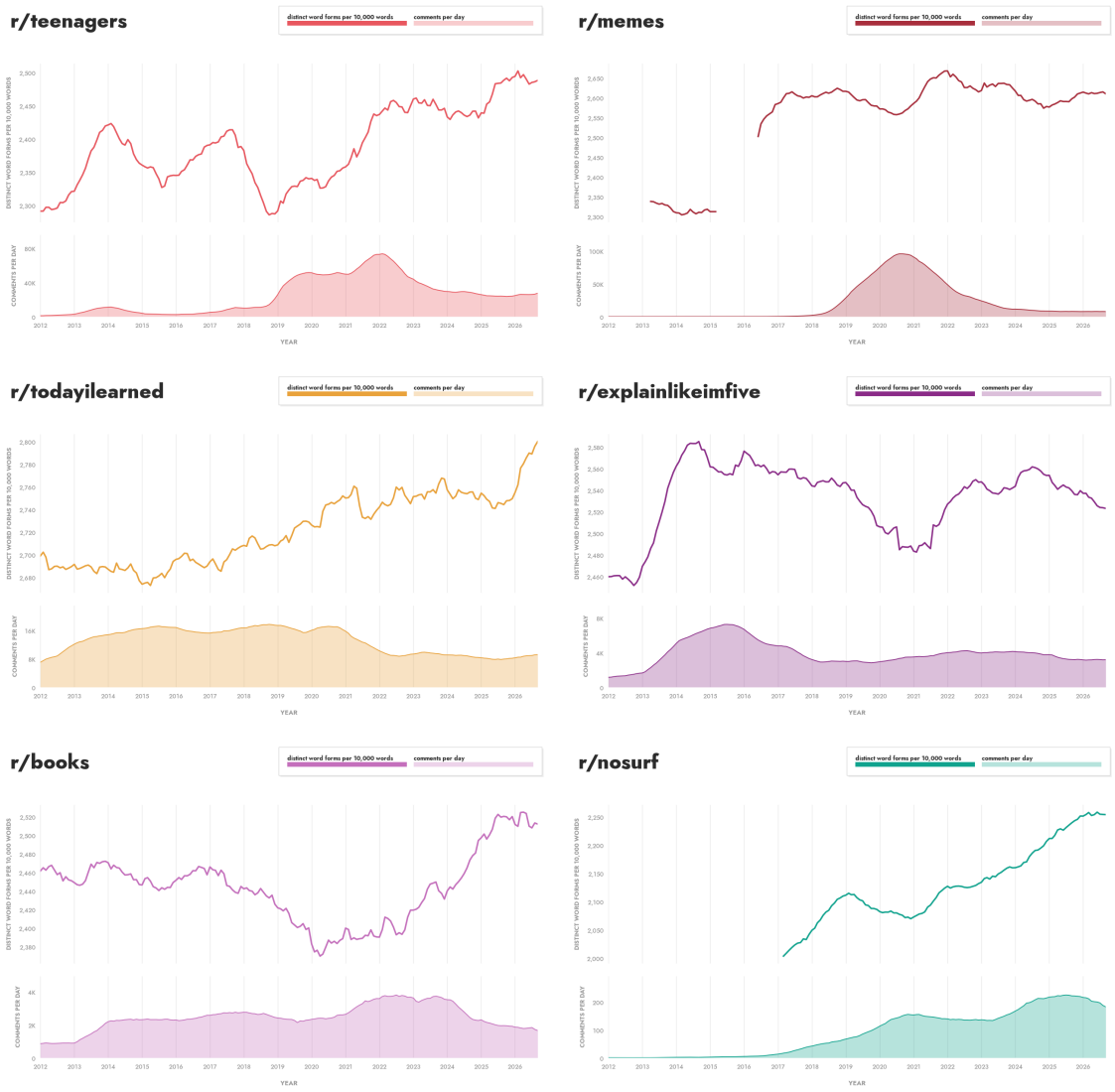
And now the same measure, but for distict word forms per 10,000 word per month per subreddit.



It is hard to trace and compare discinct word forms and disctinct lemmas between subreddits, since all subreddits cover different amount of topics and have different levels of activity. As a result, r/nosurf has less distinct lemmas, but it happens not because of actual vocabulary of contributors, but because of low activity level of this subreddit. Vice versa for r/todayilearned - since it covers a lot of topics, is has more disctinct lemmas.

Now, instead of comparing distinct lemmas between subreddits we will compare the changes in vocabulary in each subreddit separately, considering the activity level.



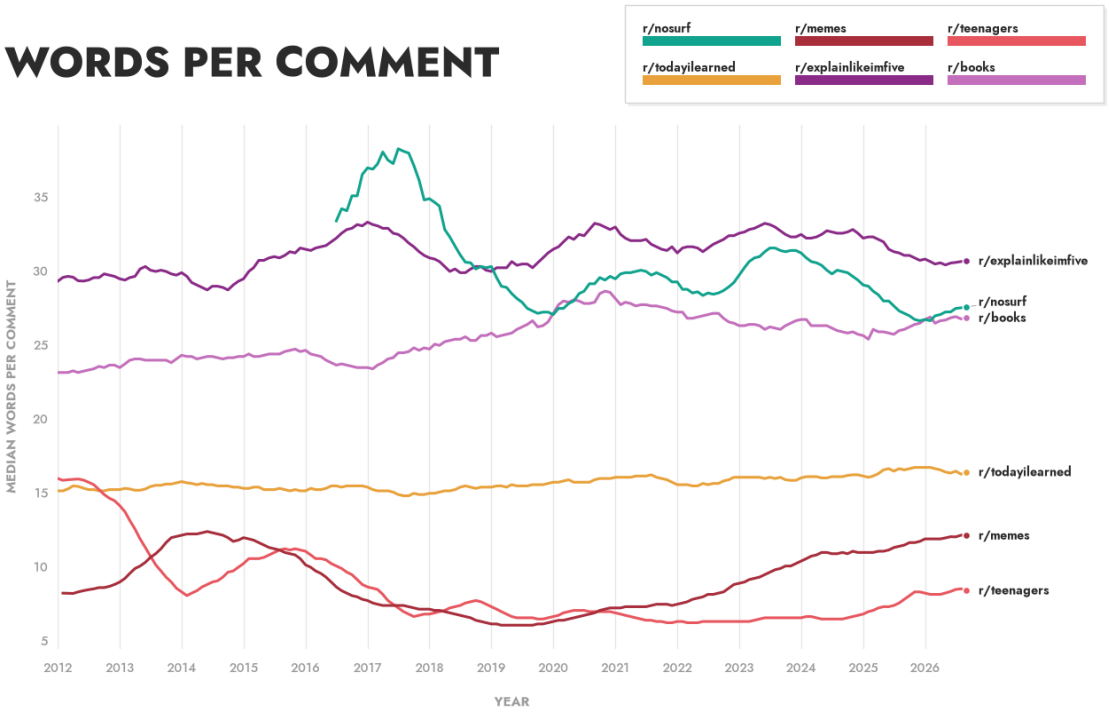


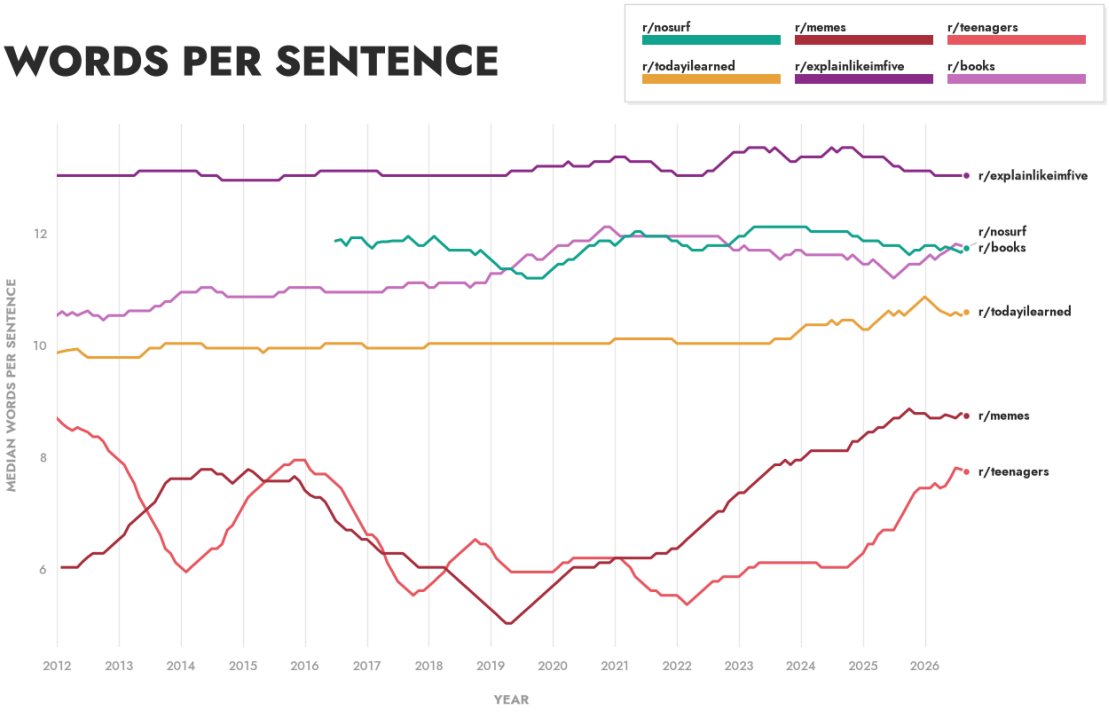
As a result, there is no noticable correlation in distinct lemmas or word forms between subreddits and with time.

1.3.2. Sentences are getting shorter

In order to check if sentences are getting shorter, we will count words per comment (1) and words per sentence (2) per subreddit per month.

Let's take a look at 12-month moving averages of each subreddit for the past 14 years. We computed median words per comment and per sentence, since it represents the average better than mean.



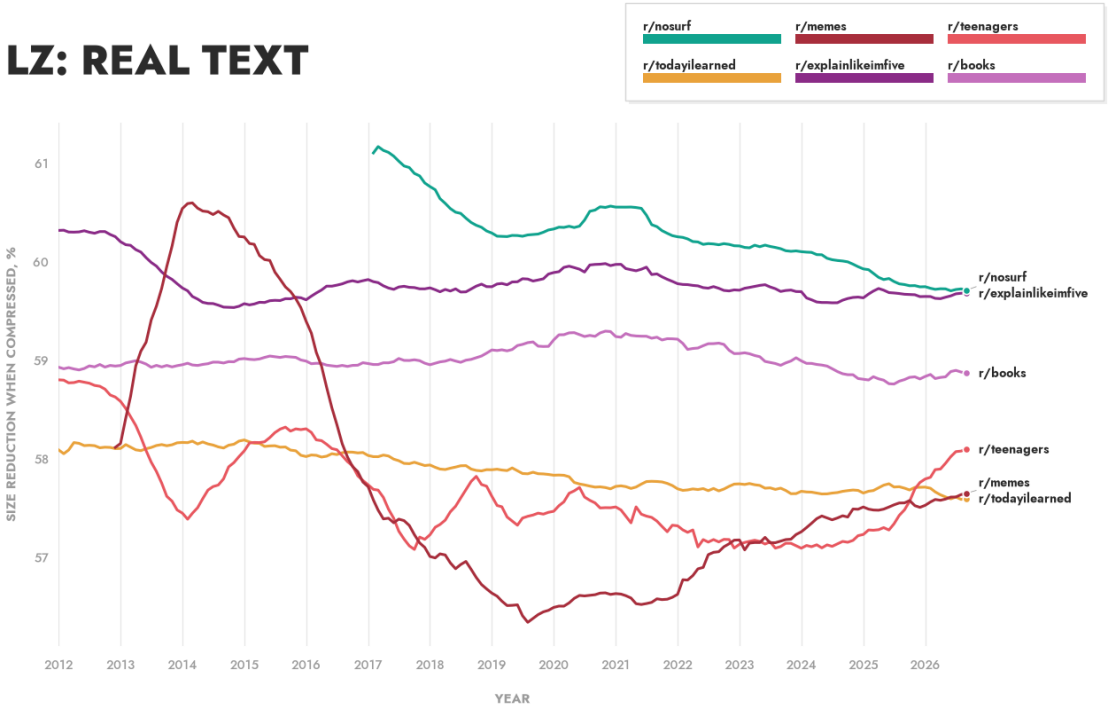


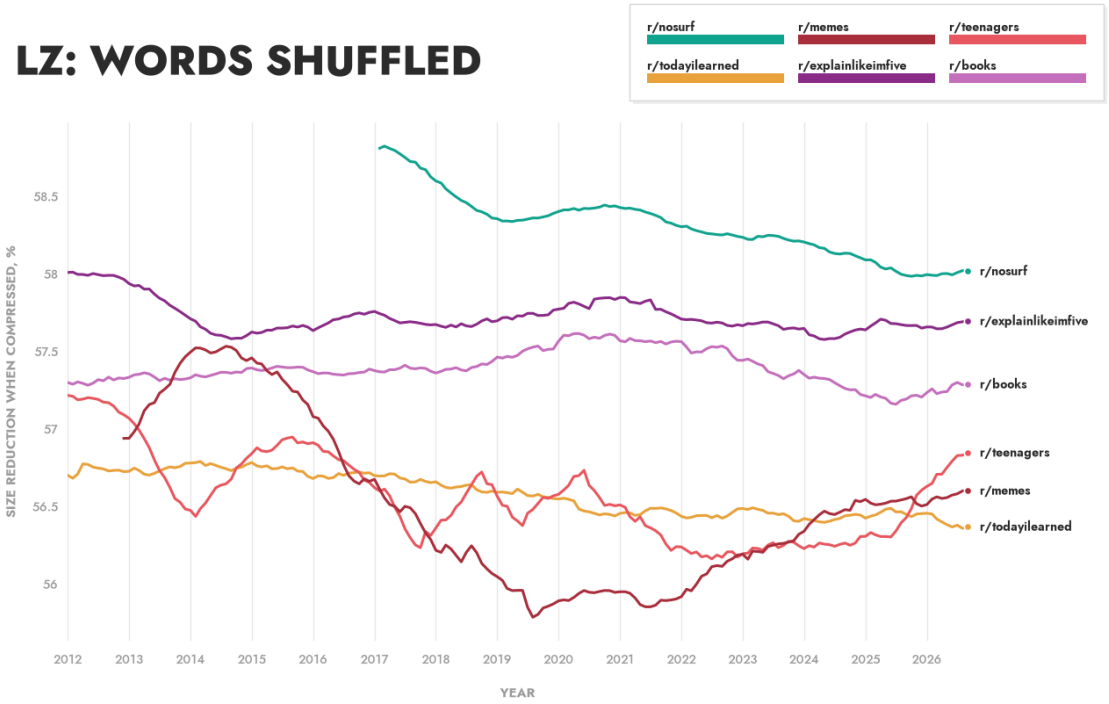
It can be seen that amount of words per comment and per sentence don't decrease over time,

however absolute amount of words per comment and per sentence are lower in short-form subreddits: r/memes and r/teenagers.

1.3.3. Lempel-Ziv algorithm

For compression we concatenate comments into fixed-size 32KB chunks per subreddit per month, compress them, get one value per subreddit per month. The value is the size reduction: 100 * (1 − compressed / original). Month value is the mean over that month's chunks. Months with fewer than 3 chunks are hidden.



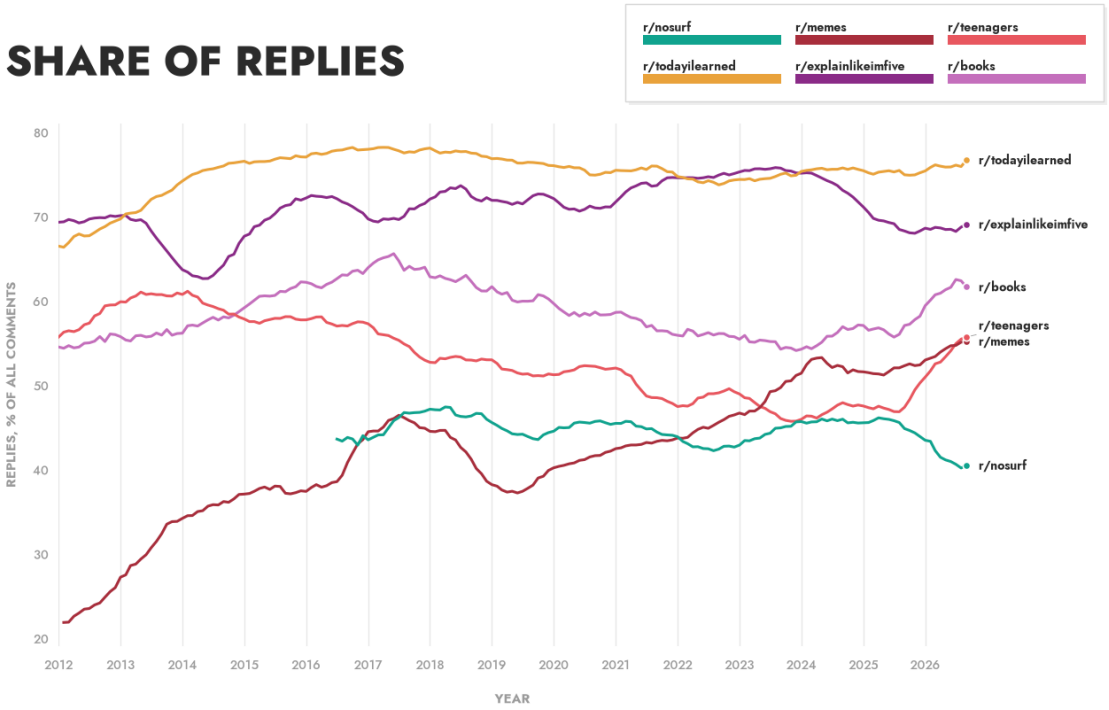


1.4. Temporal/Behavioral analysis

Let's analyze conversantions and look into:

1. Thread depth
    1. Thread size
    2. Share or replies vs top-level comments
    3. True depth

Since we have small chunks of subreddits per month, we can't measure true depth. For now the best approach is measuring the share or replies vs top-level comments. Every comment carries it's parent ID and, as a result, information if this comment replies to another comment or to post.



1.5 Result for H1

Hypothesis that "discourse is getting shorter and simpler, accelerated by the short-form platforms" is not supported. We have evidence that short-form subreddits tend to have shorter sentences and comments, but discourse is not getting shorter over time. Neither in conversantions analysis can we see the correlation. We have completed the exploration.

2 H2: Wellbeing is getting worse among doomscrollers

2.1 Description of the Analysis Concept

It usually starts harmlessly — just a few minutes of catching up on short videos. But hours later, the pattern is always the same: a subtle shift in mood, a persistent dull headache, and heavy, strained eyes. Experiencing this firsthand made me wonder: *Is this just an isolated personal reaction, or is it a quantifiable, widespread digital ailment?

To answer this, I am diving into the data to uncover the hidden behavioral patterns behind screen time and well-being. Rather than forcing disparate Kaggle datasets into a single lossy table, I am treating each as a distinct chapter in a broader story. Together, they offer a multi-faceted view of how media consumption shapes human health.

2.2 Data Landscape

There are 5 such datasets:

1. Social Media Addiction dataset

2. The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds
3. Sleep & Doomscrolling Habits Dataset
4. Social Media and Mental Health
5. Student Social Media & Mental Health

2.3 Datasets Analysis

2.4 The Conceptual Framework

To synthesize these varied dimensions, I will structure the data analysis around four core well-being clusters:

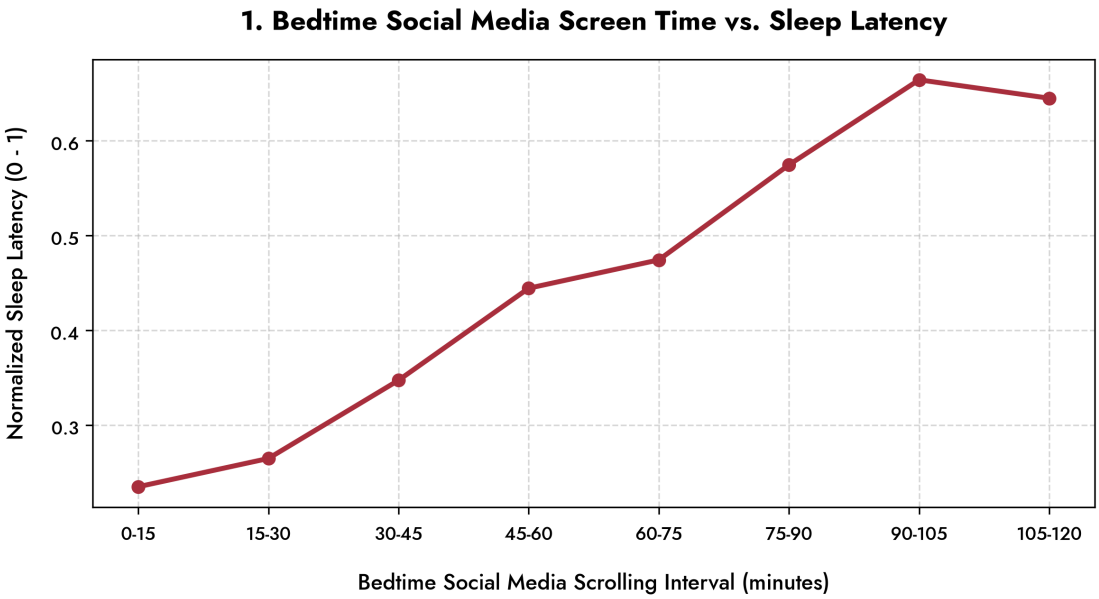
- *Sleep & Circadian Rhythms:* Mapping the impact of late-night scrolling on sleep quality and sleep delay.
- *Mental Health & Affect:* Tracking mood shifts, anxiety levels, and psychological strain.
- *Addiction & Doomscrolling Behavior:* Quantifying screen time intensity, loss of agency, and compulsion loops.

2.4.1 Sleep & Circadian Rhythms

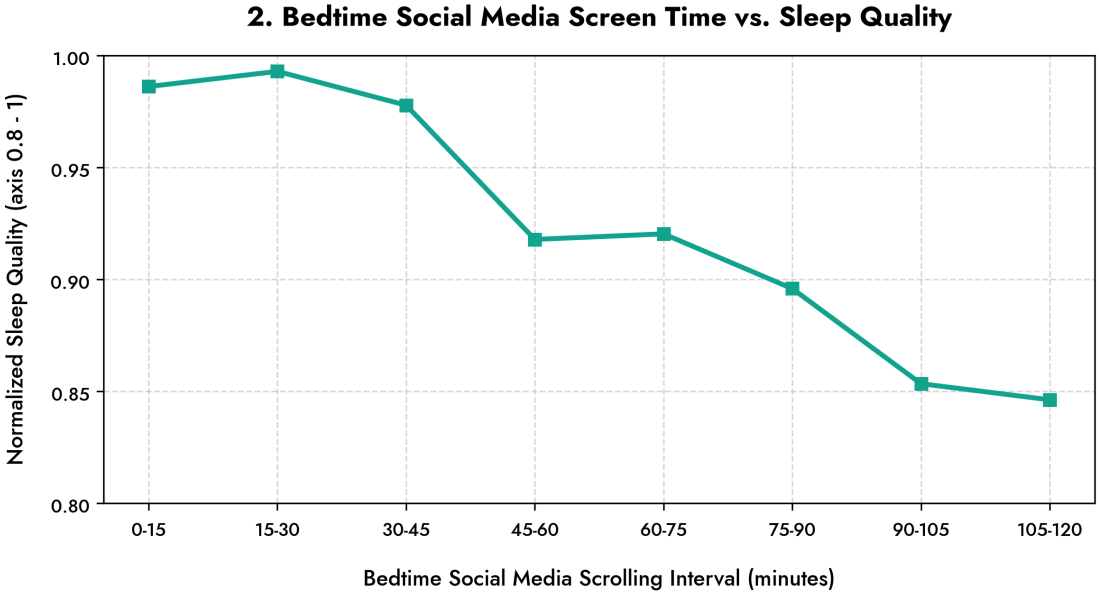
> *Picture this:* You return home exhausted after a long day with zero energy left to do anything productive. Turning to your phone for a quick scroll through lighthearted reels feels like the easiest way to decompress. But before you know it, hours pass, falling asleep becomes surprisingly difficult, your sleep is fragmented with frequent night awakenings, and you wake up feeling completely drained.
>
> It feels like a perfect vicious cycle: exhaustion drives late-night scrolling, which destroys sleep, leading to even greater fatigue the next day. But are short-form videos truly the root cause of this poor sleep quality?

*- The relationship between the amount of time spent watching videos before bed and the time it takes to fall asleep

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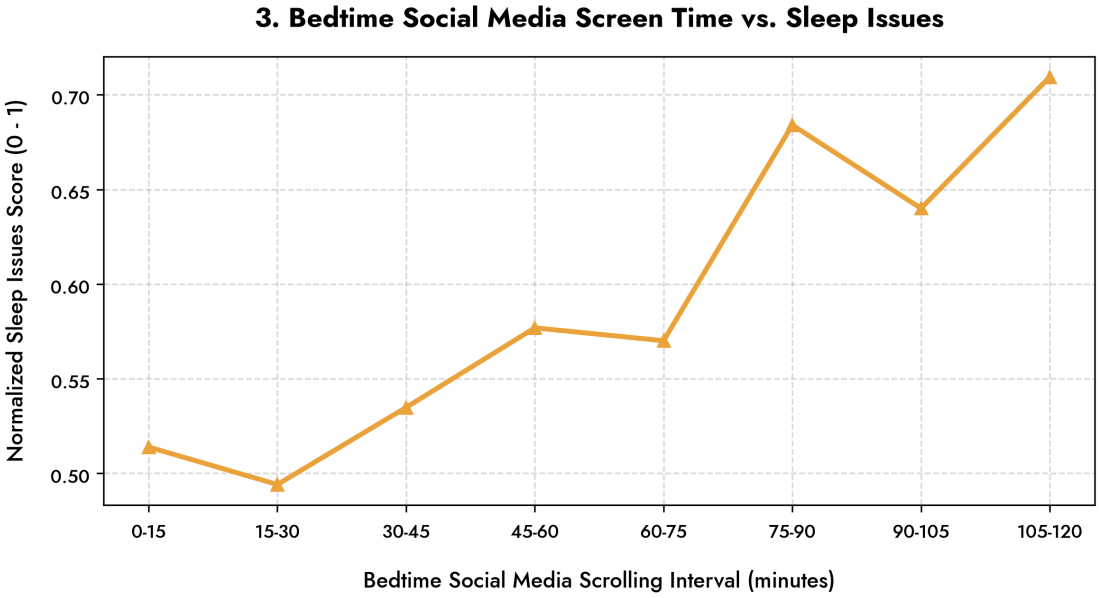


Statistical Validation: The relationship is **statistically significant** ($p < 0.001$). Spearman rank correlation shows a strong positive monotonic association ($r_s = 0.62$), meaning that respondents who scroll social media longer before bed also tend to report longer sleep latency. Since the data are cross-sectional survey responses, this is an association, not proof of a causal effect.

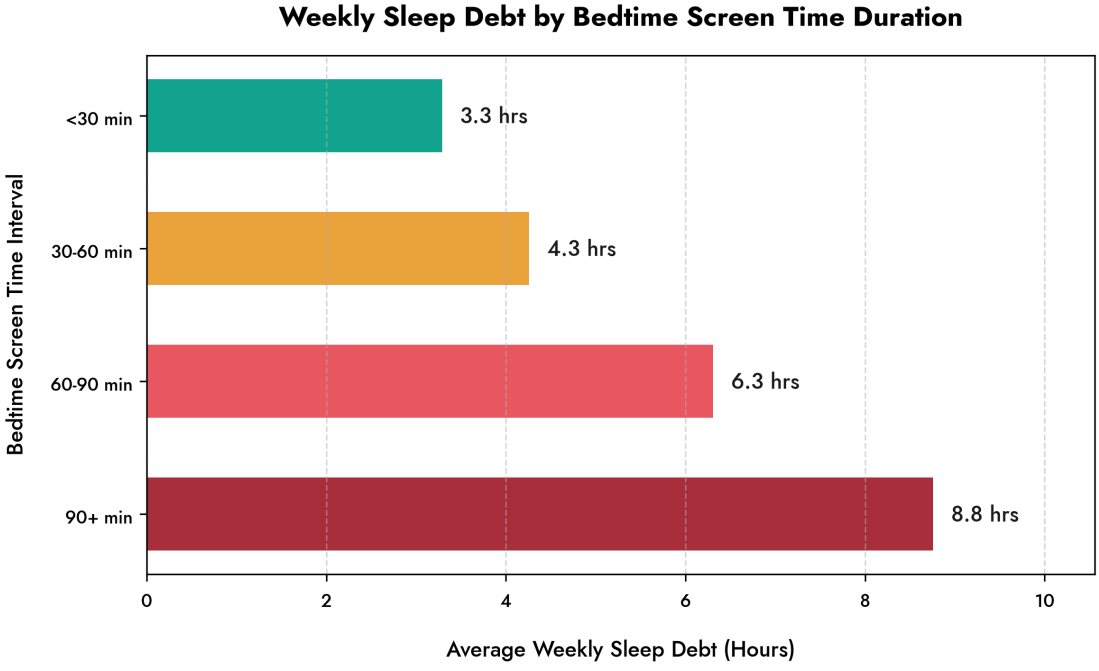


Statistical Validation: The association is **statistically significant** ($p < 0.001$) but weak to moderate: Spearman $r_s = -0.30$ ($r_s^2 \approx 0.09$). Longer bedtime scrolling is associated with slightly

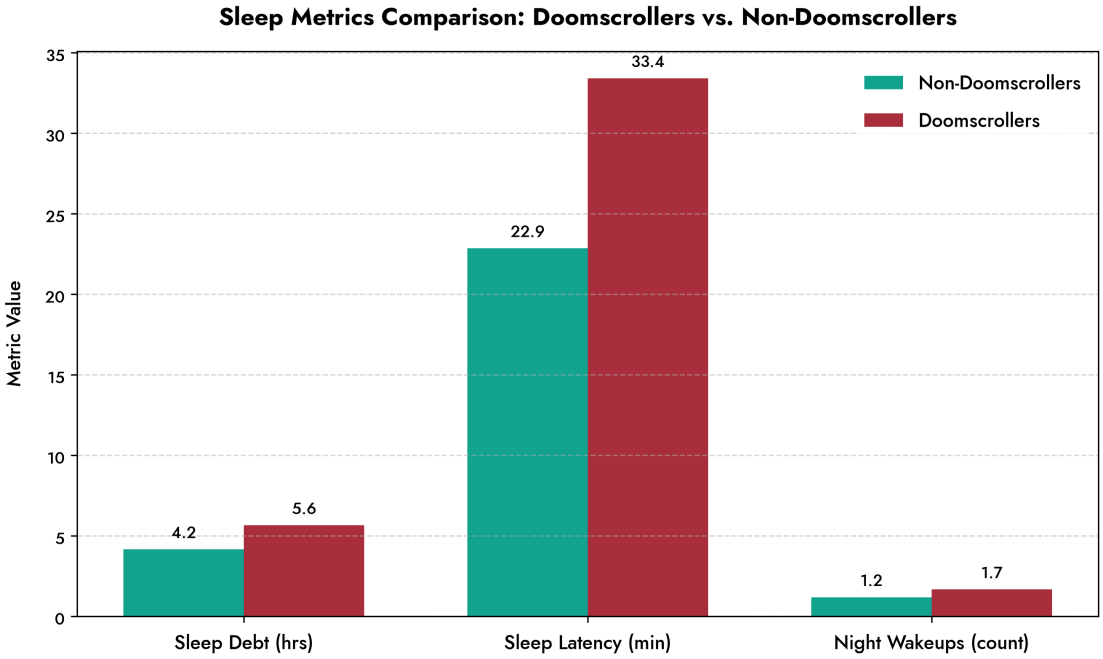
lower sleep quality.



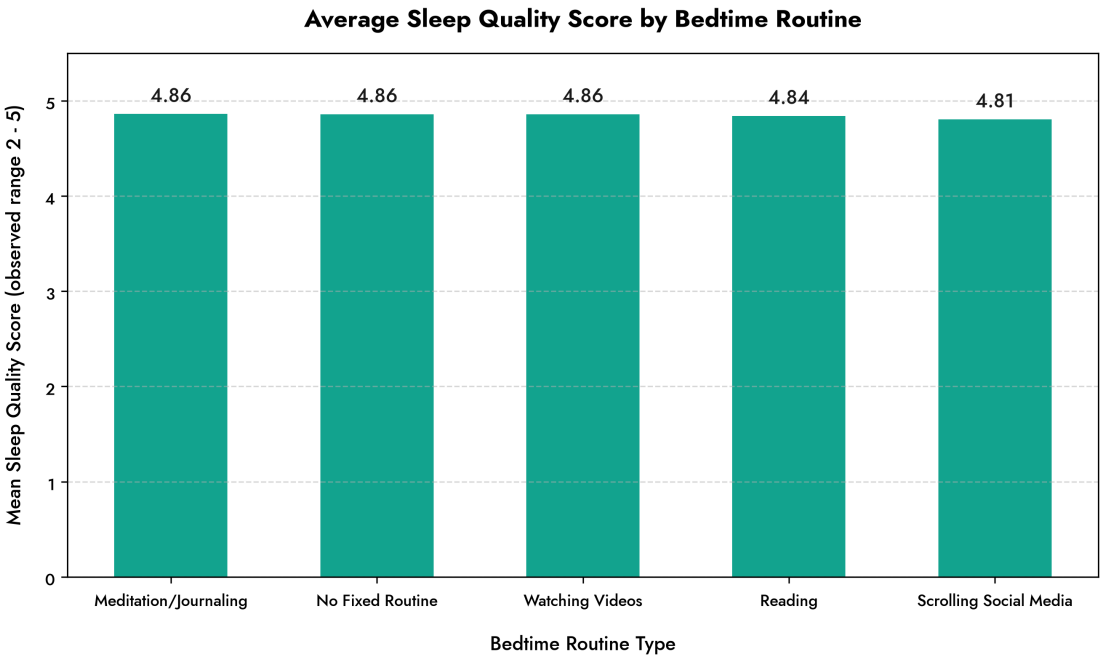
Statistical Validation: The association is **statistically significant** ($p < 0.001$) but weak: Spearman $r_s = 0.27$ ($r_s^2 \approx 0.07$). Respondents with longer pre-bedtime scrolling tend to report somewhat higher next-day fatigue. The data are correlational, so we cannot say whether scrolling causes fatigue or tired people scroll more.



Statistical Validation: The relationship is **statistically significant** ($p < 0.001$). Spearman rank correlation shows a moderate positive association ($r_s = 0.34$, $r_s^2 \approx 0.12$), and the Kruskal-Wallis test ($H = 119.44$, $p < 0.001$) confirms significant differences between screen time groups. Longer bedtime screen time is associated with higher weekly sleep debt: from about 3.3 h in the <30 min group to 8.8 h in the 90+ min group.



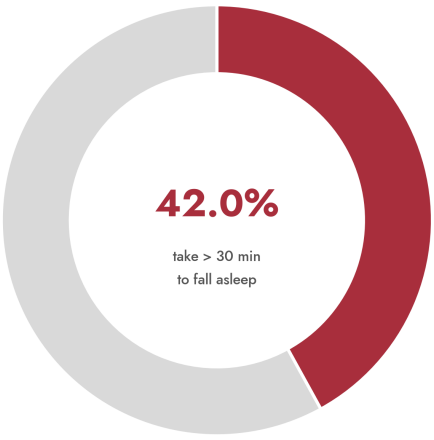
Statistical Validation: The differences between Doomscrollers and Non-Doomscrollers are **statistically significant** across all three metrics ($p < 0.001$, Mann-Whitney U test & Welch's t-test). Doomscrolling behavior is strongly associated with elevated sleep debt, significantly prolonged sleep latency, and more frequent nighttime awakenings.



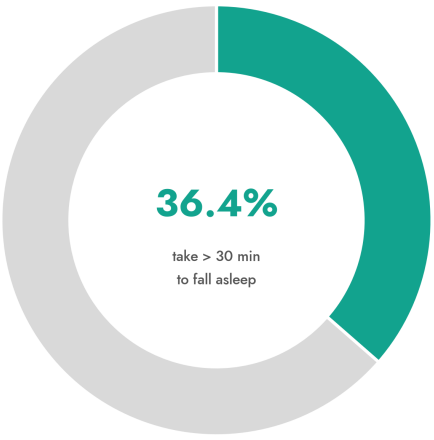
Statistical Validation: The difference in sleep quality scores across bedtime routine categories is **not statistically significant** (Kruskal-Wallis $H = 2.62$, $p = 0.623$; one-way ANOVA $p = 0.580$). We fail to reject the null hypothesis: in this data, the type of bedtime routine alone is not associated with sleep quality.

This null result should be read with caution. Because of the ceiling effect described above (87% of respondents score exactly 5), the variable has very little variance, so the test has limited ability to detect differences between groups. Combined with the significant correlations above, the data suggest that *how long* people scroll matters more than *what kind* of routine they have.

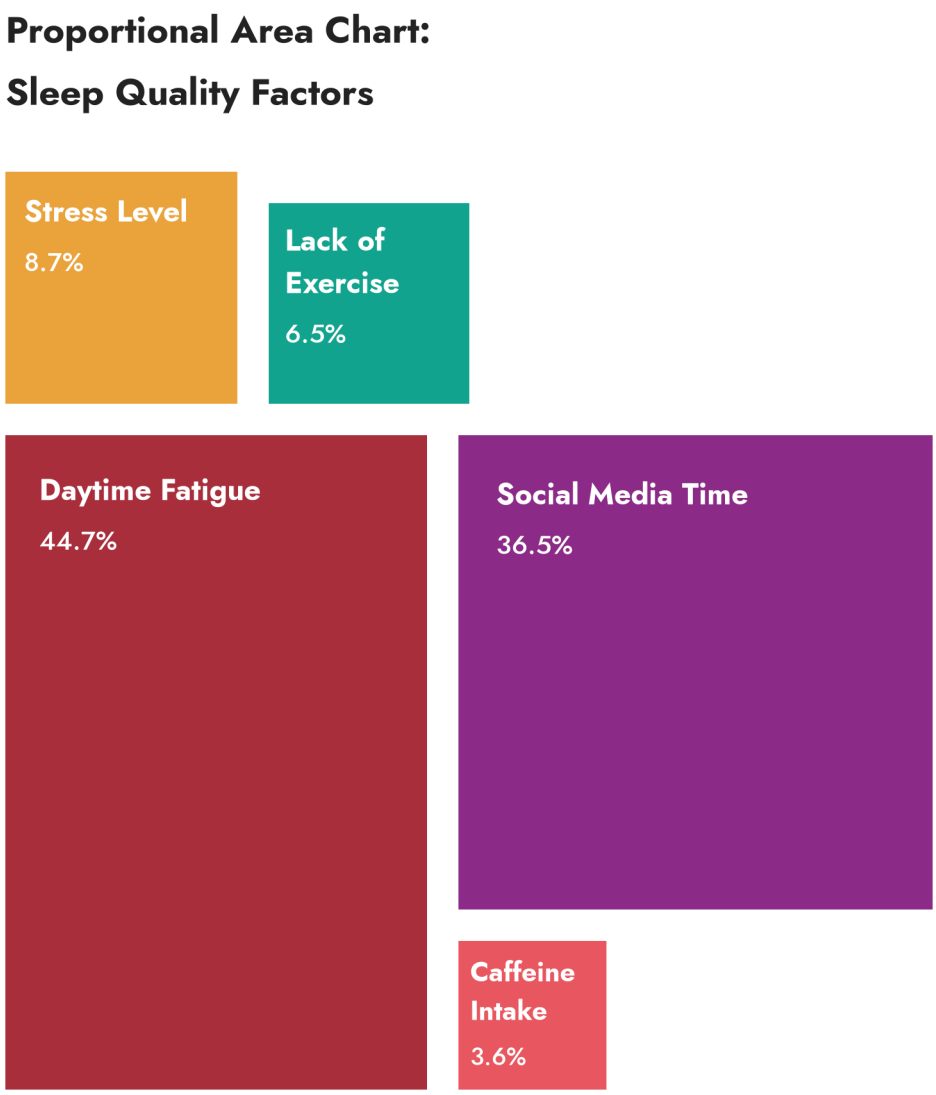
How Many People Take Longer Than 30 Minutes to Fall Asleep?



Consume negative news



Do not consume negative news



Statistical Validation:

Only bedtime social media time and daytime fatigue are significantly associated with poorer sleep quality, while caffeine intake, stress level, and lack of exercise show no significant link, so the hypothesis holds for screen time but not for the other lifestyle factors.

Summary: Does bedtime scrolling affect sleep? Yes, all three sleep indicators get worse as bedtime scrolling gets longer. All three associations are statistically significant ($p < 0.001$), but they differ a lot in strength. The table below uses the social-media scrollers subsample ($n = 382$, scrolling time < 120 min). "Change per +30 min" is the slope of a linear fit in the original units.

Indicator	Spearman r_s	r_s^2	Strength	<30 min → 90–120 min	Change per +30 min
Sleep latency (min)	**+0.62**	0.39	strong	18.3 → 43.4 min (**2.4×**)	**+9.3 min**
Daytime fatigue / sleep issues (1–10)	+0.27	0.07	weak	5.5 → 7.0 (+1.5 points)	+0.6 points
Sleep quality (2–5)	−0.30	0.09	weak–moderate	4.97 → 4.55 (−0.4 points)	−0.15 points

What this means:

1. **Falling asleep is affected the most.** Bedtime scrolling time alone explains about **39%** of the variation in sleep latency. People who scroll for 90+ minutes take, on average, about 25 minutes longer to fall asleep than those who scroll for under 30 minutes.
2. **Next-day fatigue rises, but only modestly.** Scrolling time explains about **7%** of the variation in fatigue. Moving from the lowest to the highest group adds about 1.5 points on a 10-point scale. Fatigue is the strongest single factor linked to sleep quality ($|r_s| = 0.41$), and scrolling time comes second ($|r_s| = 0.34$). Caffeine, stress and lack of exercise show no significant link ($p > 0.1$).
3. **Self-rated sleep quality drops only slightly.** Scrolling time explains about **9%** of its variation. Because of the ceiling effect (87% of respondents score 5), this indicator underestimates the real impact.

The same pattern appears in the other comparisons. Weekly sleep debt grows from 3.3 h (<30 min) to 8.8 h (90+ min), $r_s = 0.34$. Self-identified doomscrollers have more sleep debt (+1.5 h/week), take longer to fall asleep (+10.6 min) and wake up more often at night (+0.5 wakeups) than non-doomscrollers, all with $p < 0.001$. The *type* of bedtime routine on its own does not change sleep quality ($p = 0.62$). What matters is **how long** people scroll.

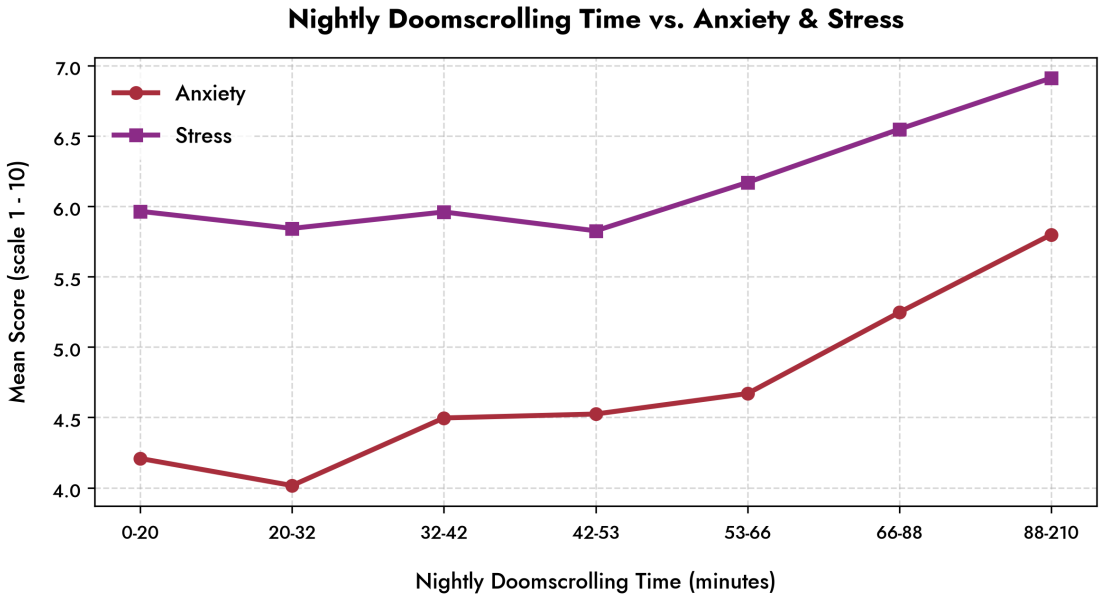
Bottom line: the data support H3 for sleep. Longer bedtime scrolling is consistently associated with worse sleep. The effect is strong for sleep latency and weak but significant for fatigue and sleep quality. These are cross-sectional, correlational survey data, so they show association, not causation. Tired people may also scroll more.

2.4.2 Mental Health & Affect

> How often do you close an app and realize you feel worse than when you opened it? Maybe a random video about some global crisis left you with a knot of anxiety you can't shake off. Maybe it was someone's flawless morning routine that made your own day suddenly look dull. Or maybe it's that familiar wave of guilt when you check the clock and see that "just five minutes" turned into an hour and a half.
>
> These feelings are easy to dismiss as passing moments, but they keep coming back, and they rarely come alone. So is there a real link between how we scroll and how we feel? Does more screen time mean more anxiety and stress, or is it the content we consume and the mindless way we consume it that does the damage?

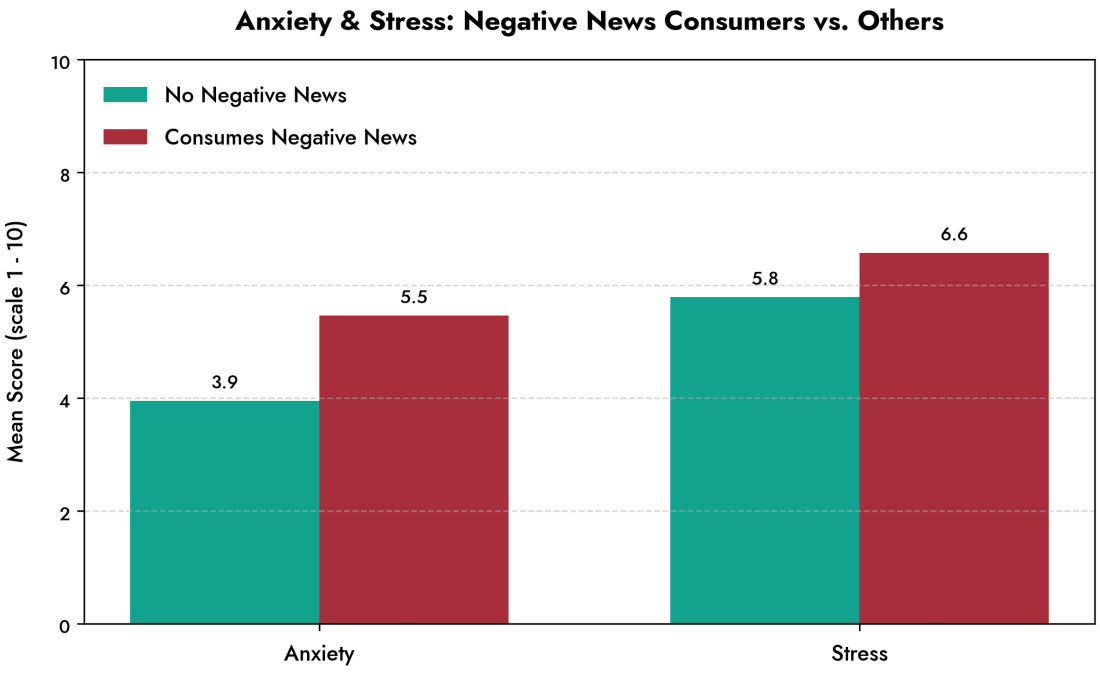
*- The relationship between nightly doomscrolling time and anxiety & stress levels

```
count    1000.000000
mean       45.923700
std        36.429492
min         0.000000
25%        20.075000
50%        42.350000
75%        66.100000
max       209.700000
Name: doomscroll_min_per_night, dtype: float64
```

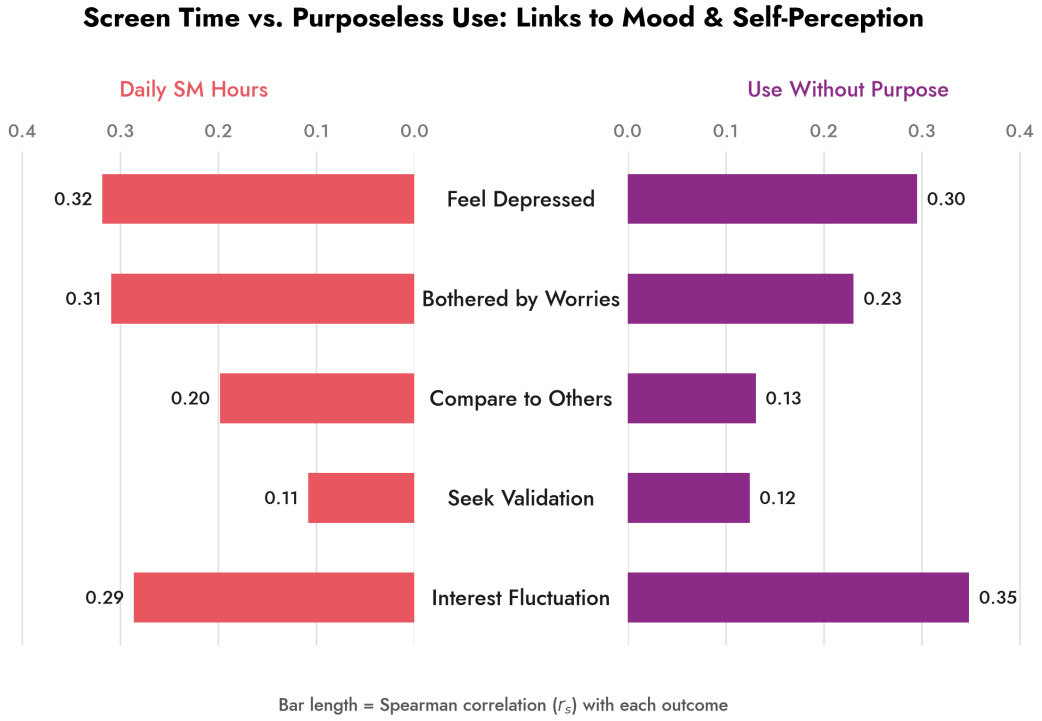


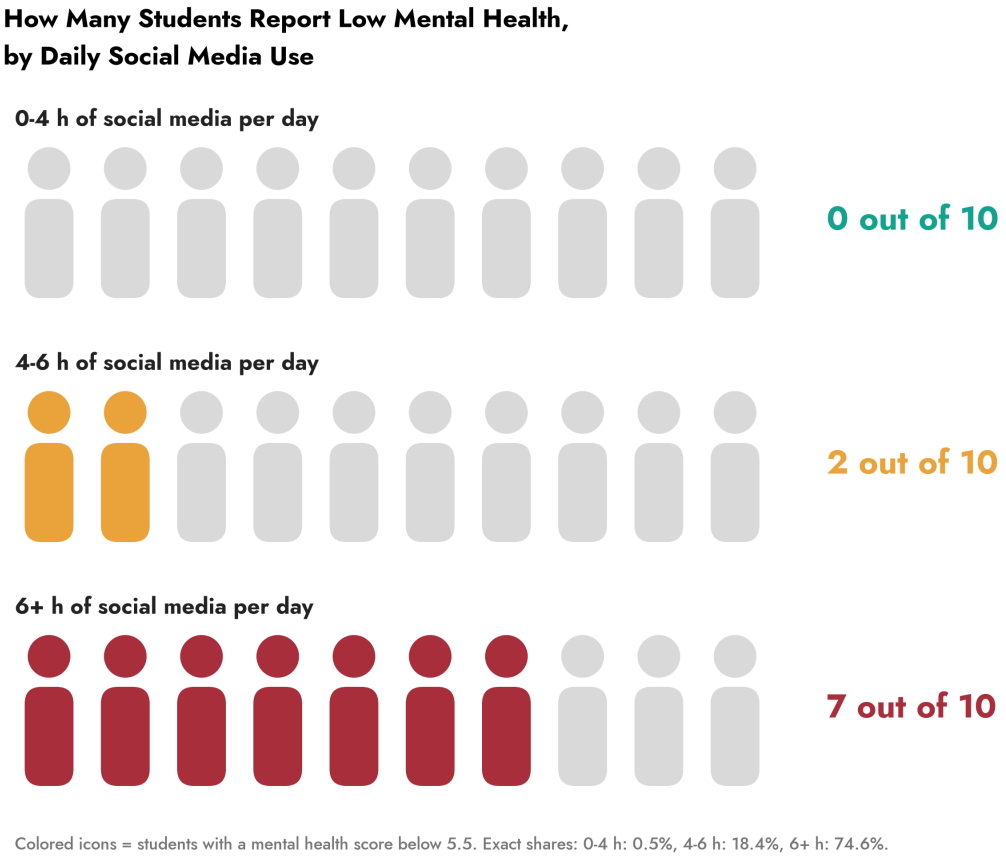
Statistical Validation: Longer nightly doomscrolling is weakly but significantly associated with higher anxiety and, to a lesser extent, stress, with the frequency of returning to the feed mattering more than the length of each session.

*- Does negative news content amplify anxiety?

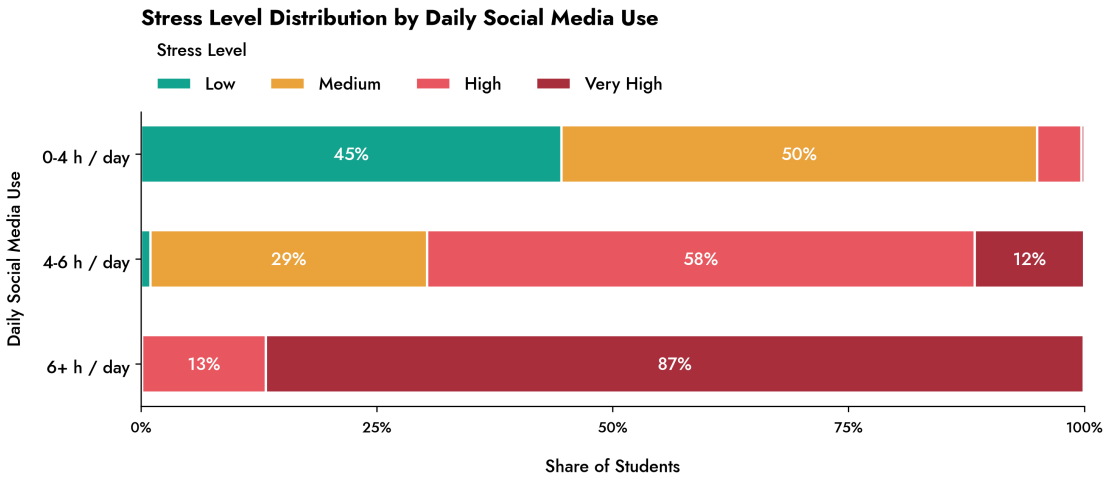


*- Screen time vs. purposeless scrolling: which predicts low mood?





*- How stress levels shift with daily social media use



Summary: Is doomscrolling linked to worse mental health?

Partly. Doomscrolling is clearly linked to anxiety and negative emotions, but not to general well-being. The data come from Doomscrolling Study 3 ($n = 502$). Doomscrolling is the mean of the 15-item Doomscrolling Scale (1–7). Respondents are split into quartiles: Q1 has DS 1.2 and Q4 has DS 2.73. "Change per +1 DS point" is the slope of a linear fit in the original units.

Indicator	Spearman r_s	r_s^2	Strength	Q1 (low) → Q4 (high)	Change per +1 DS point
Anxiety, GAD-7 (0–21)	**+0.33**	0.11	moderate	6.1 → 10.3 (**+4.2 points**)	+1.4 points
Negative affect, SPANE-N (6–30)	**+0.34**	0.11	moderate	11.8 → 15.4 (+3.6 points)	+1.3 points
Well-being, WHO-5 (0–100)	−0.06	0.00	none	46.9 → 43.9 (n.s.)	−0.1 points

What this means:

1. **Anxiety rises the most in practical terms.** Doomscrolling explains about **11%** of the variation in anxiety ($p < 0.001$; Kruskal-Wallis $H = 51.2$, $p < 0.001$). The share of respondents above the clinical threshold for moderate anxiety (GAD-7 10) grows from **19%** in the lowest quartile to **54%** in the highest, so it nearly triples.
2. **Negative emotions rise to a similar degree.** Heavy doomscrollers more often feel sad, afraid and angry. The effect size is almost the same as for anxiety ($r_s = 0.34$, $p < 0.001$). Positive emotions do not drop ($r_s = -0.05$, $p = 0.25$). Doomscrolling seems to *add* negative feelings rather than *remove* positive ones. The overall emotional balance (SPANE-B) still falls from 7.1 to 2.9 ($r_s = -0.22$, $p < 0.001$).
3. **General well-being does not change significantly.** Low well-being is common in all groups: 61% of respondents have WHO-5 50. But it hardly differs between light and heavy doomscrollers ($p = 0.17$).

Other datasets point the same way. In the Sleep & Doomscrolling dataset, doomscrollers report higher anxiety (5.1 vs 4.2 on a 1–10 scale) and stress (6.4 vs 5.9), both $p < 0.001$. In the Social Media and Mental Health dataset, daily social media hours correlate with feeling depressed ($r_s = 0.32$) and being bothered by worries ($r_s = 0.31$), both $p < 0.001$.

Bottom line: the data partly support H3 for mental health. Doomscrolling is moderately associated with higher anxiety and more negative emotions. Among heavy doomscrollers, about half reach a clinically relevant anxiety level. There is no significant link with overall well-being. The data are cross-sectional: anxious people may also doomscroll more as a way of monitoring threats.

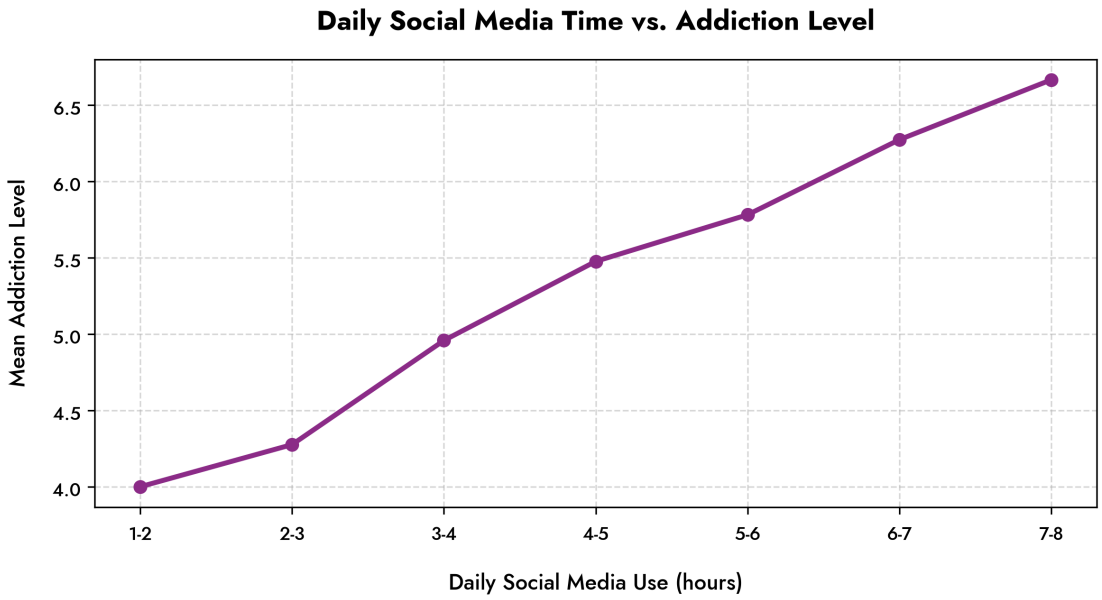
2.4.3 Addiction & Doomscrolling Behavior

In the previous sections, we saw how doomscrolling is linked to poorer sleep, higher anxiety, and lower mood. But if it leaves us feeling this way, why can't we simply stop? Most of us have promised ourselves "just one more video" or put the phone down, only to pick it up again a minute later without even noticing.

This section looks beyond the effects of scrolling and focuses on the habit itself: how screen time relates to addiction, which signs point to a loss of control, and what keeps us coming back to the feed, from the phone on the nightstand to the long gap since our last digital detox.

*- The relationship between daily social media time and addiction level

	hours_bin	mean_addiction	n
0	1-2	3.999338	151
1	2-3	4.275556	180
2	3-4	4.957895	171
3	4-5	5.476364	165
4	5-6	5.781768	181
5	6-7	6.273973	146
6	7-8	6.666341	205

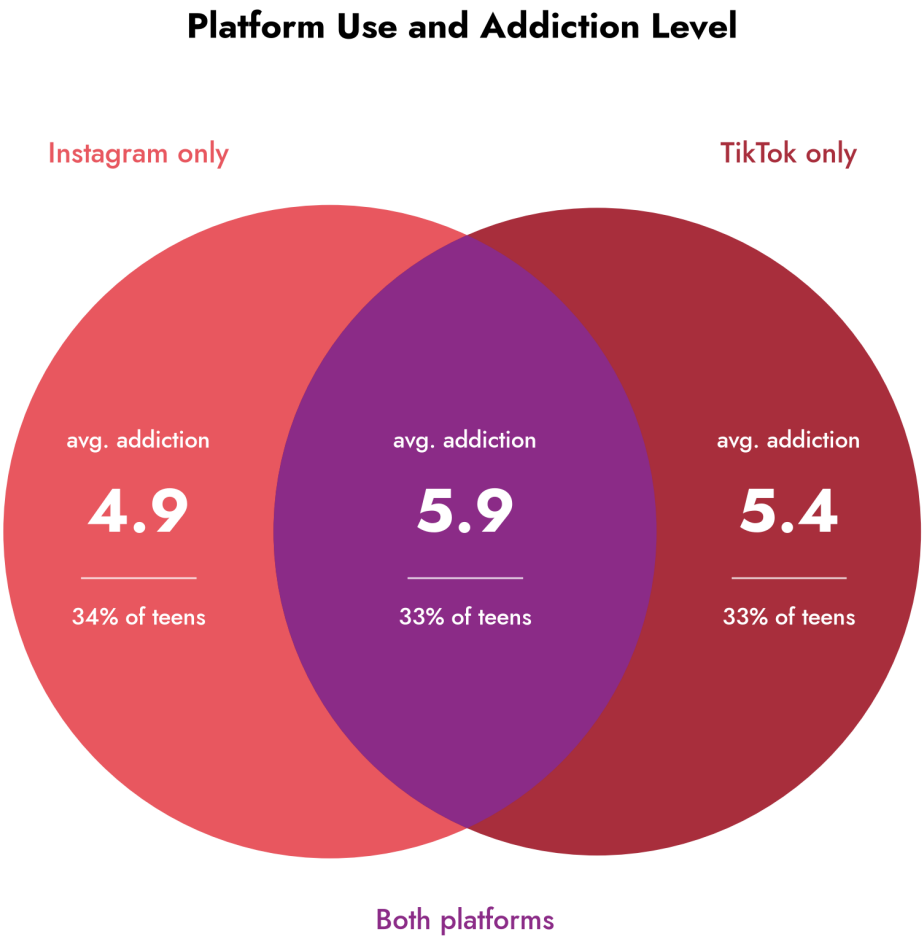


Statistical Validation: The relationship is **statistically significant** ($p < 0.001$). Spearman rank correlation shows a strong positive association ($r_s = 0.56$, $r_s^2 \approx 0.31$), and the Kruskal-Wallis test ($H = 380.53$, $p < 0.001$) confirms significant differences in addiction level between daily usage groups.

Young people who spend more time on social media tend to report substantially higher levels of addiction, with daily usage time accounting for about a third of the variation in addiction scores.

As the data are cross-sectional, this remains an association: heavy use may build addiction, but addiction may equally drive heavier use.

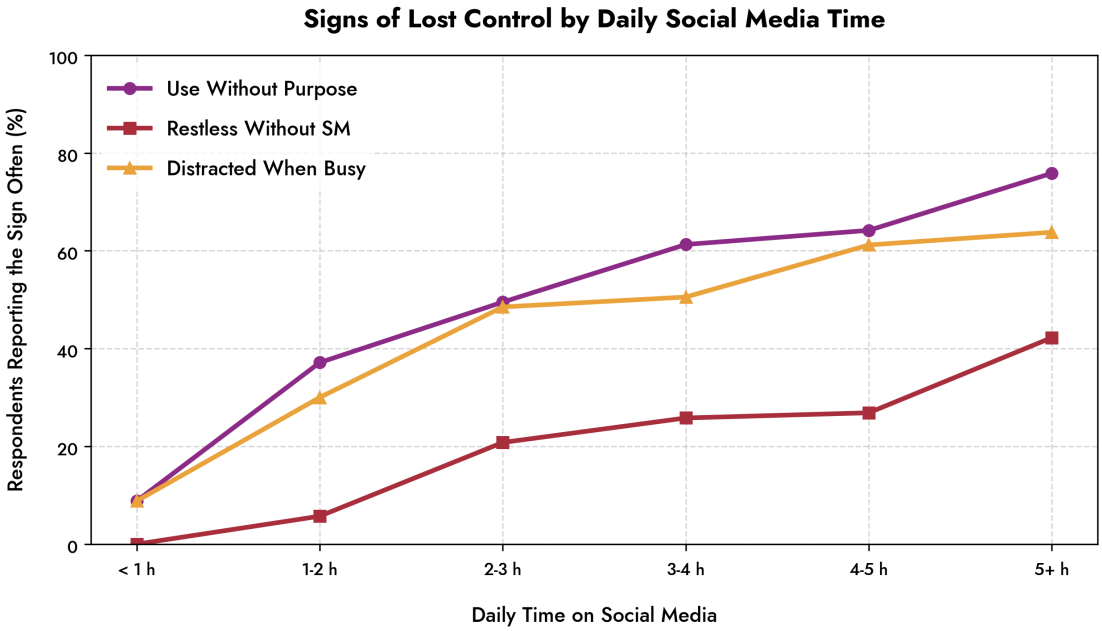
*- Which platforms are the most addictive?

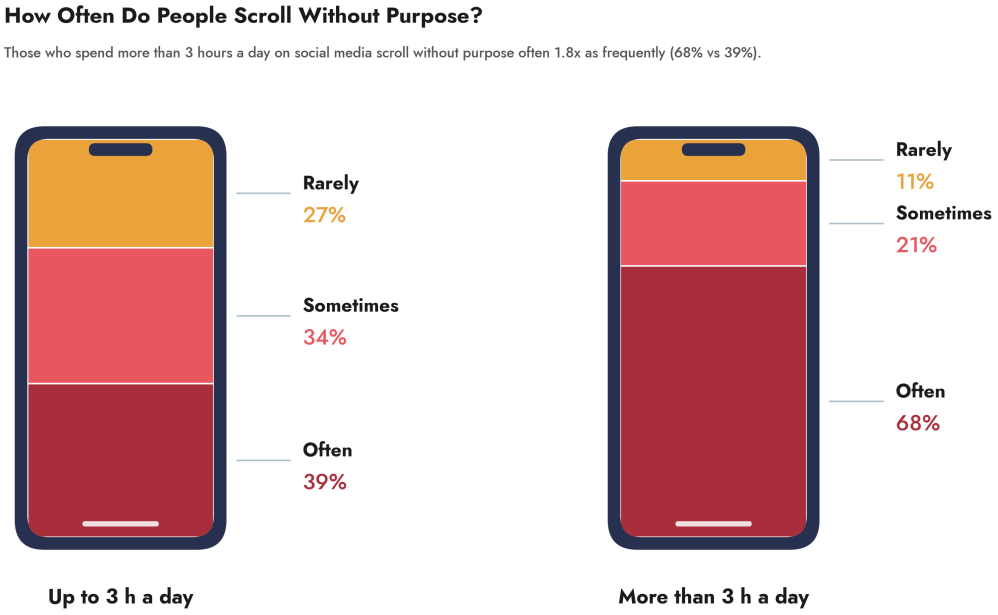


Statistical Validation:

The three groups are almost equal in size (about a third of respondents each), but their addiction levels differ **significantly** ($p < 0.001$): it is lowest among Instagram-only users (4.9), higher among TikTok-only users (5.4), and highest among those who use both platforms (5.9). Since all three groups spend nearly the same time on social media each day, this difference is not explained by longer scrolling, suggesting that using several algorithmic feeds at once is linked to stronger addiction.

*- Signs of lost control: restlessness, distraction, and aimless scrolling



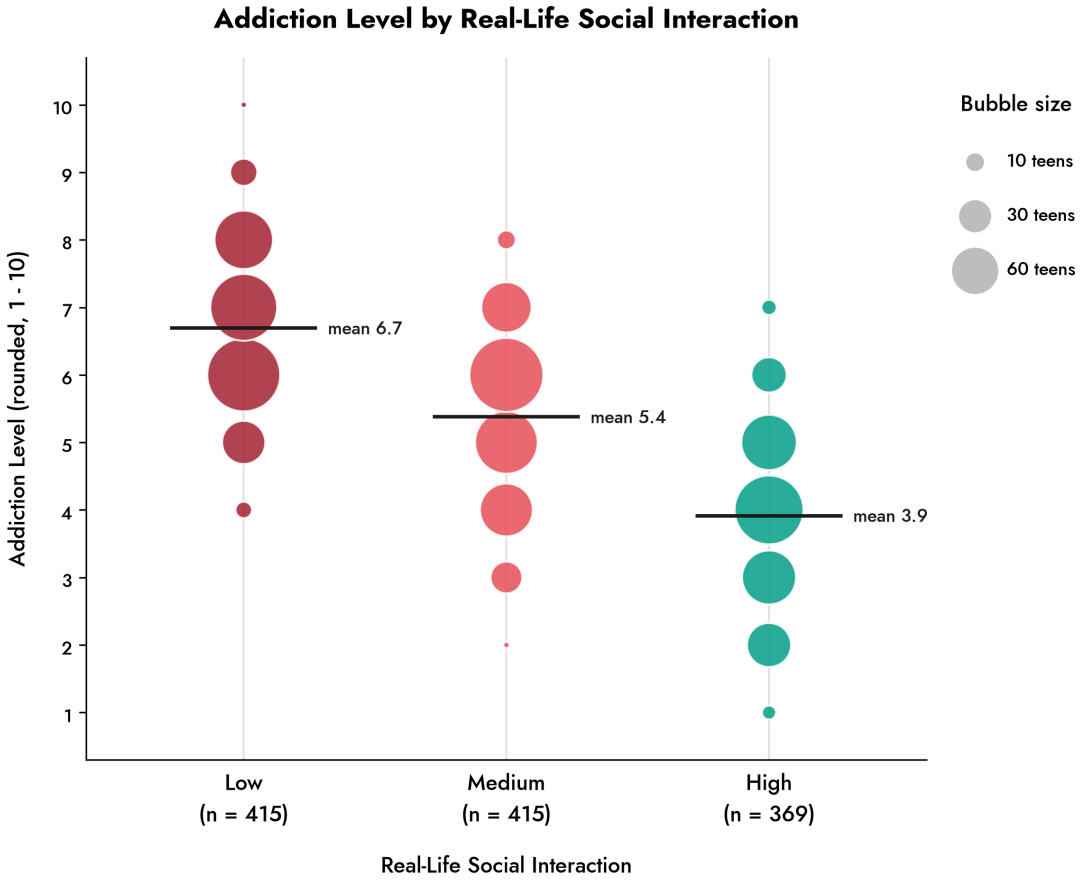


Statistical Validation:

The hypothesis is confirmed: people who spend more than 3 hours a day on social media scroll

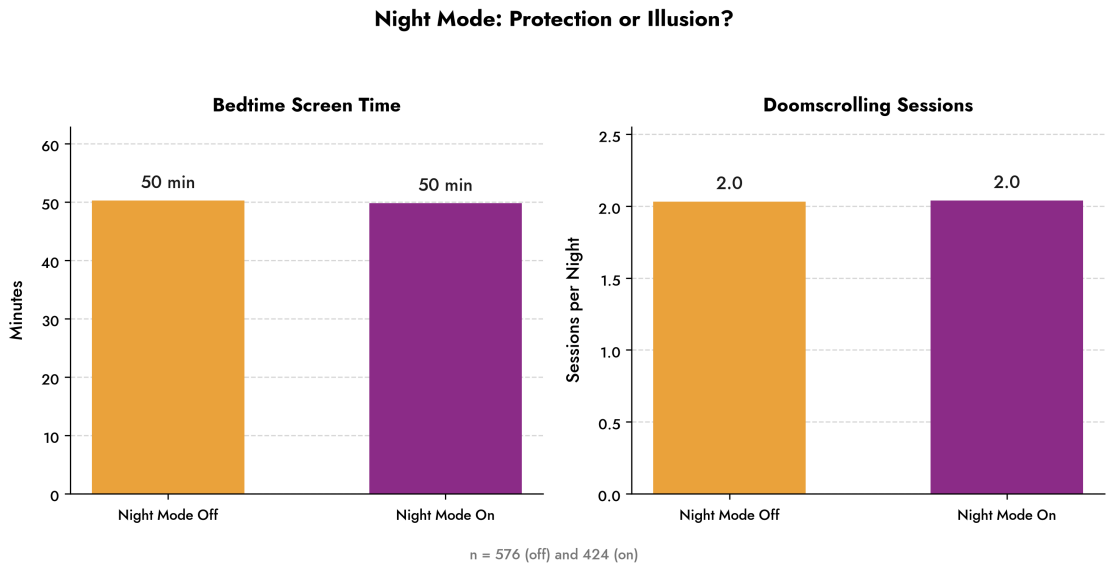
without purpose significantly more often than those who spend less, and the difference is moderate in size.

*- Addiction and social isolation: does heavy use replace real-life connection?



Statistical Validation: The hypothesis is confirmed: teens with less real-life social interaction show significantly and strongly higher social media addiction, regardless of how much time they spend online.

*- Night mode: protection or illusion?



Statistical Validation: The hypothesis is not confirmed: night mode users and non-users scroll for almost exactly the same time before bed and have the same number of doomscrolling sessions, so night mode neither protects against late-night scrolling nor encourages it.

Summary: Is doomscrolling a sign of compulsive social media use? Yes. Doomscrolling is most strongly linked to addiction-like use, more weakly to screen time and only slightly to loss of self-control. The data come from Doomscrolling Study 3 ($n = 502$; for hours, $n = 498$, excluding 4 answers above 24 h/day). Doomscrolling quartiles are the same as in 2.4.2.

Indicator	Spearman r_s	r_s^2	Strength	Q1 (low) → Q4 (high)	Change per +1 DS point
Social media addiction, BSMAS (18–90)	**+0.60**	0.36	strong	29.4 → 52.2 (**+22.7 points**)	**+7.1 points**
Daily social media time (h/day)	+0.22	0.05	weak	3.4 → 4.1 h (+0.7 h)	+0.3 h
Self-control, BSCS (1–5)	−0.17	0.03	weak	3.03 → 2.71 (−0.3 points)	−0.1 points

What this means:

1. **Compulsion is the main companion of doomscrolling.** Doomscrolling explains about **36%** of the variation in social media addiction scores ($p < 0.001$; Kruskal-Wallis $H = 171.9$). This is the strongest link in the whole H3 analysis. Heavy doomscrollers score about **77%**

higher than light ones. Related measures confirm this: problematic internet use (GPIUS-2, $r_s = 0.51$) and fear of missing out (FOMO, $r_s = 0.44$), both $p < 0.001$.
2. **Screen time grows, but modestly.** Heavy doomscrollers spend about 40 minutes more per day on social media ($r_s = 0.22$, $p < 0.001$). Doomscrolling is less about *how many* hours people spend and more about *how* they use them: compulsively and without control.
3. **Self-control drops only slightly.** The link is significant but weak ($r_s = -0.17$, $p < 0.001$). Doomscrolling explains only about **3%** of the variation in self-control. So loss of agency here appears to be specific to social media rather than a general personality trait.

Other datasets show the same pattern. In the Sleep & Doomscrolling dataset, doomscrollers have about 1 h more daily screen time (4.2 vs 3.2 h) and check their phone more often at night (2.8 vs 2.2 times). In the Social Media Addiction dataset, daily hours correlate strongly with addiction level ($r_s = 0.56$, $p < 0.001$). In the Social Media and Mental Health dataset, heavier users feel more restless without social media ($r_s = 0.39$) and more often use it without a purpose ($r_s = 0.39$).

Bottom line: the data strongly support the idea that doomscrolling is part of an addiction-like use pattern. Its strongest link is with compulsive social media use ($r_s = 0.60$). Its links with screen time and general self-control are weaker. As before, these are correlations from survey data, not proof of causality.

3 H3: Crises intensify doomscrolling and negativity

3.1 Overview

This section will cover topic of H3 "crises intensify doomscrolling and negativity".

What we currently have:

3.1.1 Events

Four major events fall inside the data range(brainrot era). We checked their dates on 2026-10-04 against news and reference sources.

1. The WHO declares COVID-19 a pandemic (11 March 2020)
2. The full-scale Russian invasion of Ukraine (24 February 2022)
3. The Hamas attack on Israel that starts the Gaza war (7 October 2023)
4. The assassination of Charlie Kirk (10 September 2025)

The last one was not on our list at first. It showed up as the busiest r/nosurf day: 11 September 2025 had 530 comments, 2.8 times the month before. Three threads carried 72% of that day, and the two biggest were about the assassination.

3.1.2 Communities

1. Short-form: r/memes + r/teenagers
2. Long-form: r/books + r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

We only test whether non-news communities react. Later adding news subreddits is considered.

3.1.3 Data

Comments from the six subreddits, January 2012 to September 2026. We use cleaned comments only: bots, moderator messages, templated text, removed text and spam are dropped.

3.2 Methods

List of methods we used for data exctraction:

1. Using mean VADER compound score (tone)
2. Searching words about the event itself, to check that the event reached the community
3. Searching for crisis words in general
4. `doomscroll*` mentions
5. Brainrot words (rizz, skibidi, gyatt, brain rot, tung tung, mewing)
6. Length of posting sessions

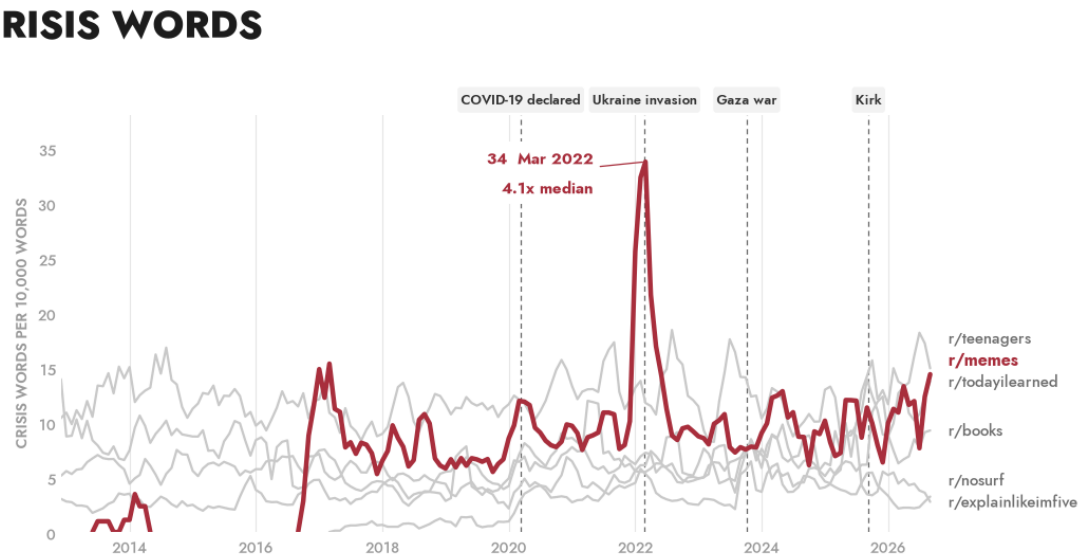
3.3 Do crises show up in what people write?

Crisis talk spikes in r/memes

What we did: Counted a broad list of crisis words per 10,000 words. The list: war, pandemic, covid, lockdown, invasion, Ukraine, Gaza, Hamas, Israel, Palestine, genocide, terror, shooting, assassination, crisis, recession, famine, refugees, earthquake, hurricane. The chart shows one line per subreddit on one scale, as a 3-month mean.

Result: r/memes shows one clear crisis peak: 34 crisis words per 10,000 words in March 2022, 4.1 times its monthly median (8.2). The other communities have no common peak:

- r/teenagers peaks in July 2026 (18.2, median 4.2)
- r/books in December 2025 (12.3, median 5.9)
- r/todayilearned in August 2022 (18.4, median 11.6)
- r/nosurf in December 2020 (6.8, median 3.1)
- r/explainlikeimfive in October 2012 (14.8, median 5.4)



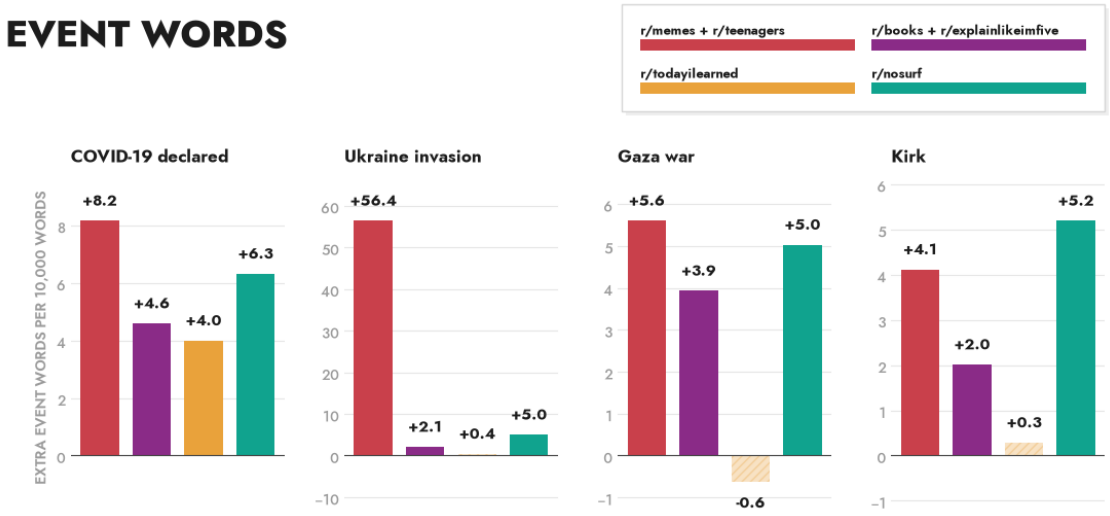
3.4 Did each event reach each community?

Nearly every event reached nearly every community

What we did: Counted event-specific words per 10,000 words in the 4 weeks after each event and compared them with the 8 weeks before. The words are covid/pandemic/lockdown, Ukraine/Russia/Putin/invasion, Gaza/Hamas/Israel/Palestine and Kirk/Turning Point/assassination. To judge a change we repeated the same calculation at 400 random fake dates. The share of fake dates with an equal or larger change is p. The chart shows one panel per event, and solid bars are beyond random dates ($p < 0.05$).

Result: Event words rose in 13 of 16 community-event pairs ($p < 0.05$ against random dates). The change per 10,000 words:

- Short-form: +8.2 (COVID), +56.4 (Ukraine), +5.6 (Gaza), +4.1 (Kirk)
- Long-form: +4.6, +2.1, +3.9, +2.0
- r/nosurf: +6.3, +5.0, +5.0, +5.2
- r/todayilearned: +4.0 for COVID, but not for the other three (+0.4, -0.6, +0.3)



3.5 Did tone get more negative after the events?

A small dip in tone, too weak to call a finding

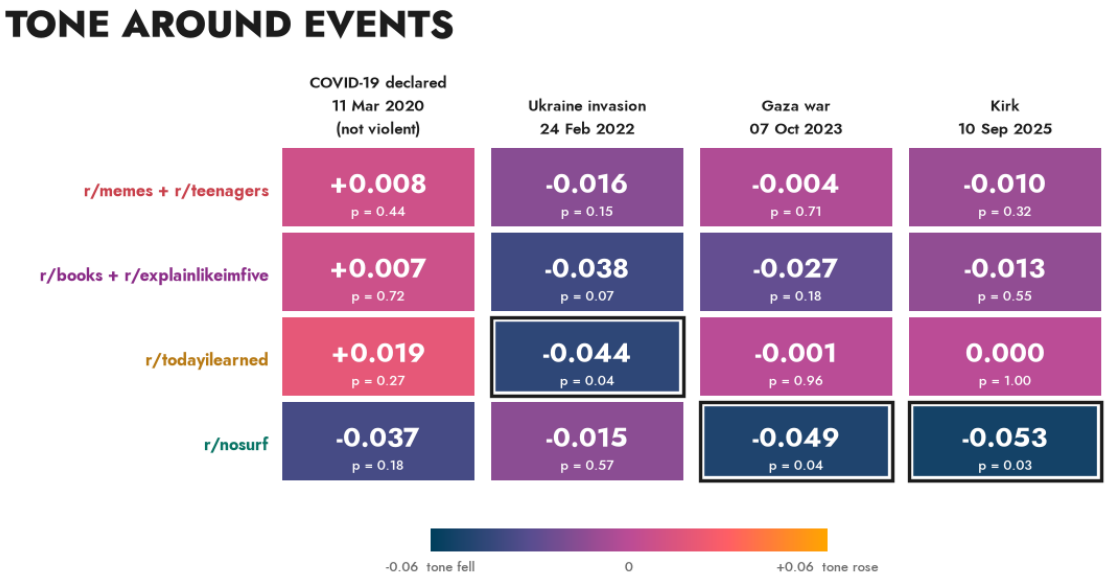
What we did: Measured the change in mean VADER compound score (-1 to +1) in the 4 weeks after each event versus the 8 weeks before. The tile matrix shows four community groups and four events: dark blue = tone fell, amber = tone rose, magenta = no change. A ring marks tiles beyond the placebo range ($p < 0.05$), and each tile shows its p-value.

Result: After the three violent events (Ukraine, Gaza, Kirk) tone was lower in 11 of the 12 community-event pairs, by up to 0.05 compound points. The twelfth, r/todayilearned after Kirk,

is zero. The largest drops:

- r/nosurf: Kirk -0.053, Gaza -0.050
- r/todayilearned: Ukraine -0.044
- long-form: Ukraine -0.038

After the COVID declaration tone rose slightly in three groups (+0.007 to +0.019) and fell in r/nosurf (-0.037). Three of the 16 tiles are ringed: r/nosurf after Gaza ($p = 0.04$) and after Kirk ($p = 0.03$), and r/todayilearned after Ukraine ($p = 0.04$). About 0.8 would be expected by chance, so we do not call this a finding. For scale, r/nosurf's mean tone fell by 0.20 over nine years (0.36 in 2017, 0.16 in 2026).



3.6 Did talk about doomscrolling rise after the events?

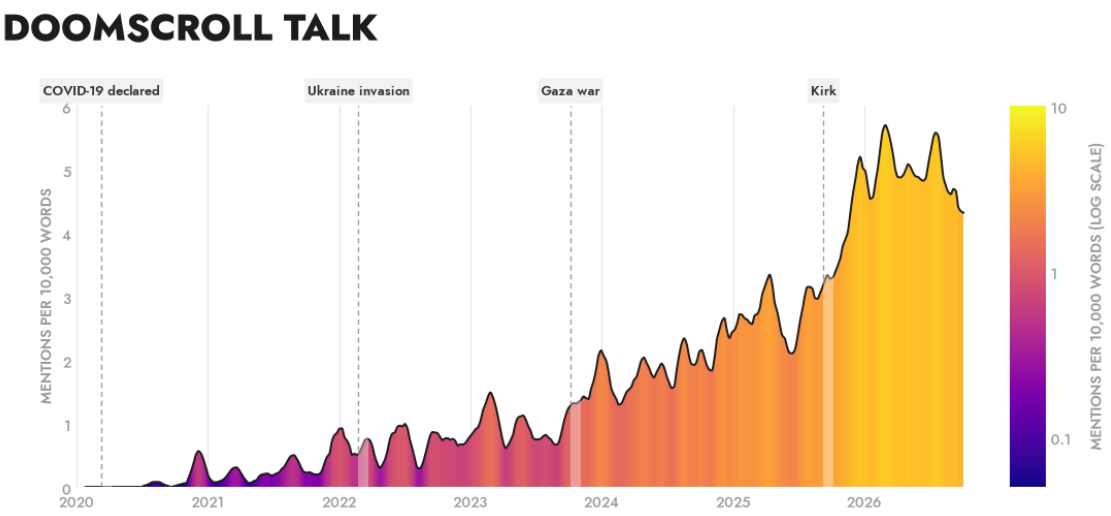
Doomscroll talk climbs, with or without crises

What we did: Counted `doomscroll*` and `doom surf*` mentions per 10,000 words in r/nosurf only, where we hold every comment. The rate is weekly, as a 4-week rolling rate from 2020. The chart is smoothed once more with a 5-week moving average, but the numbers below come from the unsmoothed rate. The events are dashed lines, and the 4 test weeks after each are pale bands.

Result: The rate climbs steadily. Per 10,000 words it is:

- 0.08 in 2020, 0.29 in 2021, 0.66 in 2022, 1.14 in 2023
- 1.88 in 2024, 3.24 in 2025, 4.95 in 2026 (to September)

The 4-week rate peaked at 6.3 in the week of 19 July 2026 and is 3.8 now. The four-week changes at the events are small blips on that rise: COVID 0.0 ($p = 1.0$), Ukraine +0.8 ($p = 0.05$), Gaza +0.5 ($p = 0.16$), Kirk -0.3 ($p = 0.28$), against a random-date spread of about 0.3. So the events do not explain the climb.



3.7 Did brainrot words spike around the crises?

Are brainrot spikes really tied to major events?

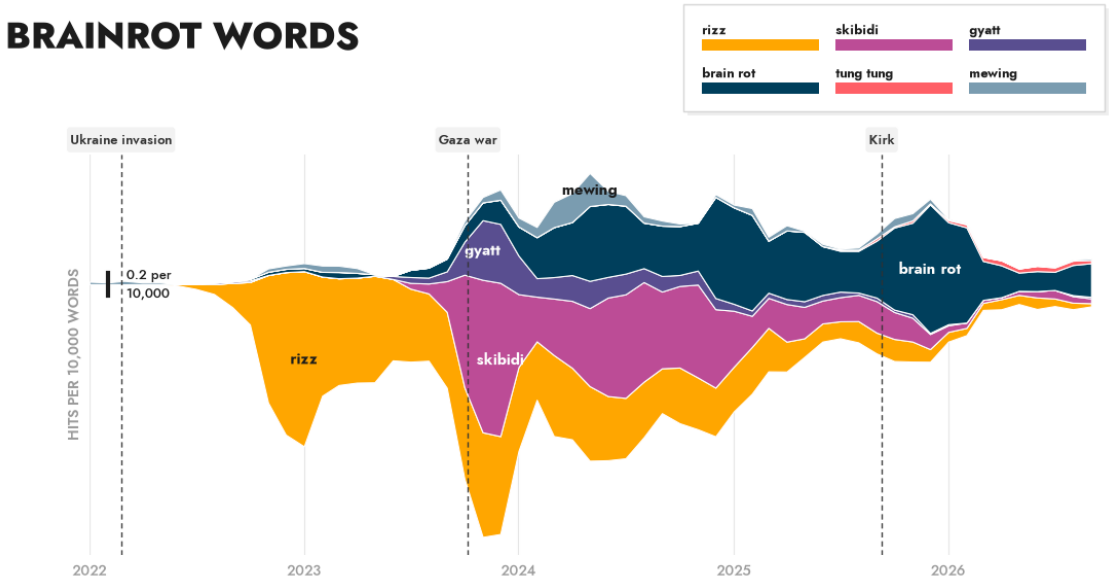
What we did: Counted brainrot words per 10,000 words in r/memes + r/teenagers (monthly, 3-month mean) and drew one band per word, thickness = rate. The axis starts in January 2022, because COVID-19 (March 2020) came before the slang existed. Then we pooled all the words and compared the 8 weeks before each event with the weeks after it, against random dates (every day from November 2022 that is more than 120 days from an event).

Result: In all six subreddits together, rizz is 83% of brainrot-word hits in 2022. Skibidi (23%) and gyatt (9%) join in 2023, and 'brain rot' / 'brainrot' take 41% in 2024 and 72% in 2025. The pooled rate in r/memes + r/teenagers, per 10,000 words:

- Gaza war, 4 weeks after: 0.95 before, 1.53 after (+0.58, p = 0.41), no jump yet
- Gaza war, 8 weeks after: 0.95 before, 3.40 after (+2.44), beyond every random date (they give -1.41 to +1.41)
- Kirk, 8 weeks after: 0.96 before, 1.11 after (+0.16, p = 0.69)

The jump comes late: the weekly rate is 4.45, 11.00 and 5.36 in weeks 5, 6 and 7 after 7 October 2023, against 0.43 to 1.65 in the 8 weeks before. We chose the 8-week window after seeing this weekly series, so read the p-value as a guide. With all six subreddits pooled the Gaza change is smaller (+0.38, p = 0.035). Week 6 rests on 25 sampled comments with brainrot words, from 24 authors on 3 days. The comments are not about the war: of 91 comments with brainrot words in November 2023, none mentions Gaza, Israel, Palestine or Hamas.

So brainrot words do jump after the Gaza war, five to seven weeks late and not about the war. We cannot tell whether the crisis or the slang's own rise drove it. After Kirk's killing pretty much nothing moves.



3.8 Did posting sessions in r/nosurf get longer?

Posting sessions barely move, with one lead after Ukraine

What we did: Built gap-based sessions (messages by one author with gaps of at most 30 minutes) from comments and submissions, r/nosurf only. Kept sessions with 2 or more messages and split them into five length buckets (1 minute, 2-4, 5-9, 10-29, 30+). Bars show the share of sessions in each bucket: light outlined bars are the 8 weeks before, solid bars the 4 weeks after. As a yardstick we repeated the largest bucket shift at 100 random fake dates.

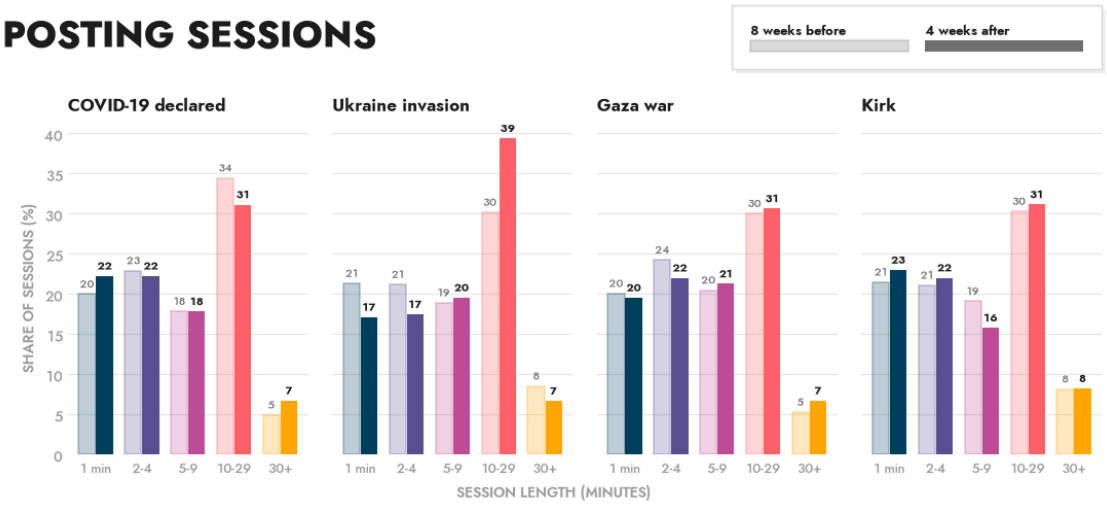
Result: The share of sessions with two or more messages, before vs after:

- COVID: 11.5% vs 11.4%
- Ukraine: 13.9% vs 11.9%
- Gaza: 12.5% vs 13.3%
- Kirk: 17.0% vs 16.0%

Around COVID, Gaza and Kirk no length bucket moves by more than 3.3 percentage points, and the median session stays at 6 to 7 minutes.

After the Russian invasion of Ukraine the 10-29 minute bucket grows from 30% to 39% of the sessions with two or more messages (+9.2 points), while 1-minute and 2-4-minute sessions shrink (21% to 17% each). That shift is beyond what random dates produce ($p = 0.02$). With four events tested, one p-value of 0.02 or lower happens about 8% of the time by chance, so it is a lead, not a conclusion.

So sessions are unchanged at COVID, Gaza and Kirk. Only the Ukraine invasion moves them.



3.9 Do the same r/nosurf authors write differently after an event?

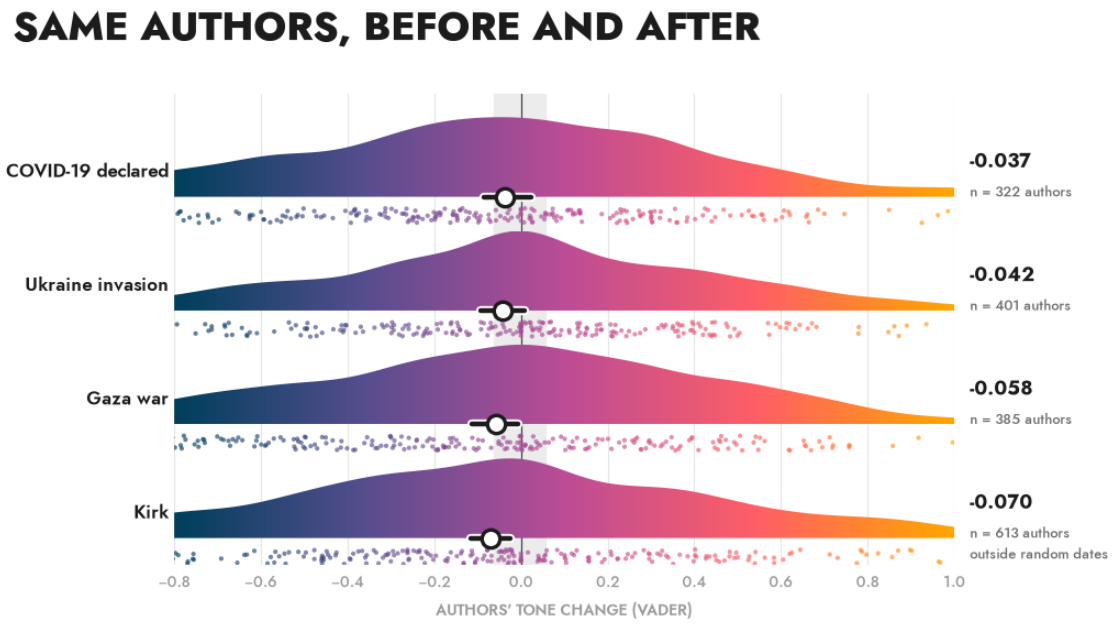
The same authors write less positively

What we did: Took authors with comments in both windows (8 weeks before, 4 weeks after). Each dot is one author's change in mean VADER score, and the cloud shows the same data as a distribution. The big dot and bar are the mean with a 95% bootstrap interval, and the gray band is the 95% range of the same statistic at 120 random fake dates.

Result: The same authors wrote less positively after all four events:

- COVID: -0.037 (322 authors)
- Ukraine: -0.042 (401)
- Gaza: -0.058 (385)
- Kirk: -0.070 (613)

The intervals of Gaza and Kirk exclude zero. Only the Kirk change lies outside the random-date band (-0.063 to +0.058). This is the tone of the text, not the mood of the people.



3.10 Result

Let's pull the threads together:

- **Events reach the communities.** Event words rose in 13 of 16 community-event pairs. r/todayilearned is the exception (4.4).
- **Tone: small, consistent direction, weak evidence.** Tone was lower after the three violent events in 11 of 12 pairs, by at most 0.05. Three of 16 tests are outside the placebo range, about what we expect by chance given many correlated tests (4.5). The same r/nosurf authors wrote 0.04-0.07 less positively afterwards, with only Kirk beyond random dates (4.9).
- **Doomscroll talk climbs steadily, and the events do not explain it.** In r/nosurf the rate went from 0.08 per 10,000 words in 2020 to 4.95 in 2026. The event effects are small blips (4.6).
- **Brainrot words jump weeks after the Gaza war, not after Kirk.** The pooled rate in r/memes + r/teenagers rises from 0.95 to 3.40 per 10,000 words in the 8 weeks after the Gaza war, peaking in week 6, beyond random dates. After Kirk's killing it does not move (0.96 to 1.11). The jump comes late and is not about the war, so we cannot say the crisis caused it (4.7).
- **Posting sessions** are unchanged at three events. After the Ukraine invasion r/nosurf sessions shifted towards 10-29 minutes (30% to 39% of sessions with 2+ messages, $p = 0.02$), one of four events (4.8).

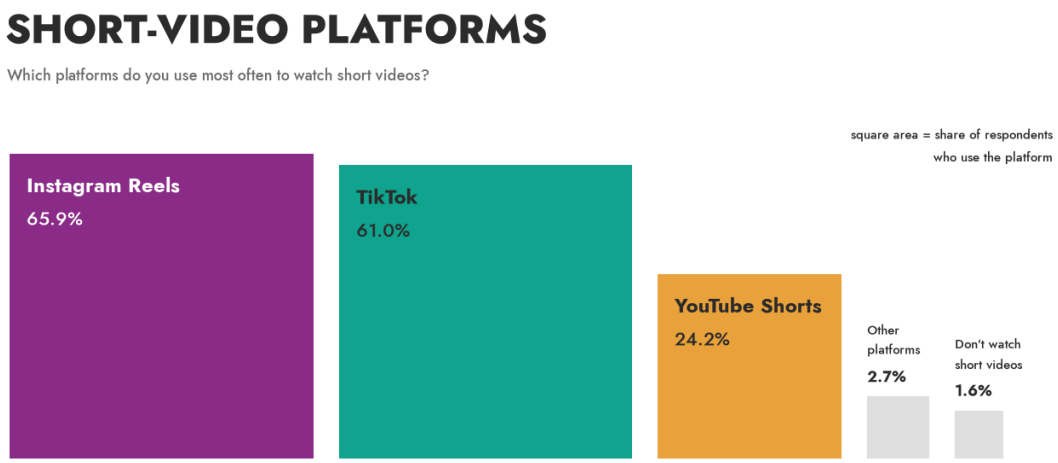
Verdict: H4 is only weakly supported. The crises do reach the communities and tone dips a little afterwards, but too weakly to call a finding. Doomscroll talk climbs steadily, with or without crises. Brainrot words jump after the Gaza war, late and not connected to event itself.

4 Questionnaire

We are exploring not only the effects of the doomscrolling on human brains, but also we trying to solve this problem and investigate what causes people to doomscroll against their will and how to break this addiction.

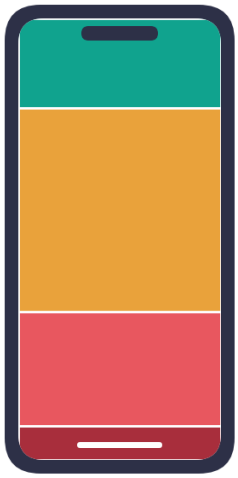
We have collected 182 answers to our questionnaire and discovered the following: 1. 87.2% of people want to stop scrolling 2. 80.4% of people think that doomscrolling is caused by coping mechanism 3. 14.4% of people think that their communication skills are degrading because of doomscrolling - aligns with our fail to prove hypothesis 1 4. 69.9% of people think that habit of doomscrolling is the main challenge to stop watching short-form format

Below is the complete result of the questionnaire



DAILY WATCH TIME

On average, how much time per day do you spend watching short videos?



Less than 1 hour
20.3%

1–2 hours
46.2%

2–4 hours
25.8%

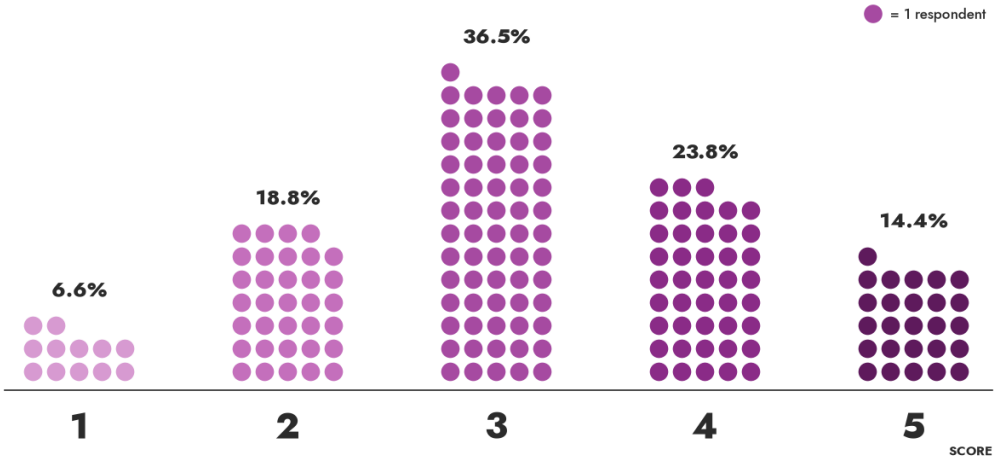
More than 4 hours
7.7%

79.7%
watch short videos for more than an hour a day

33.5%
for more than two hours a day

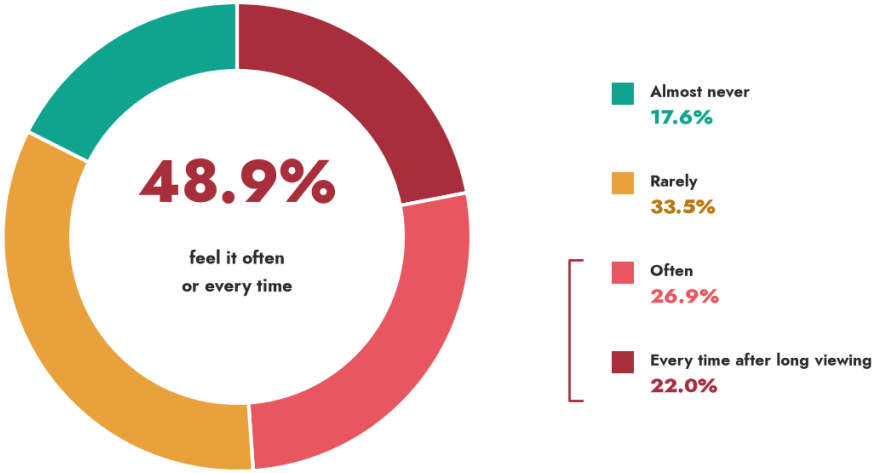
FOCUS ON LONG FORMATS

How has your ability to focus on long formats (books, lectures, videos over 15 min) changed?



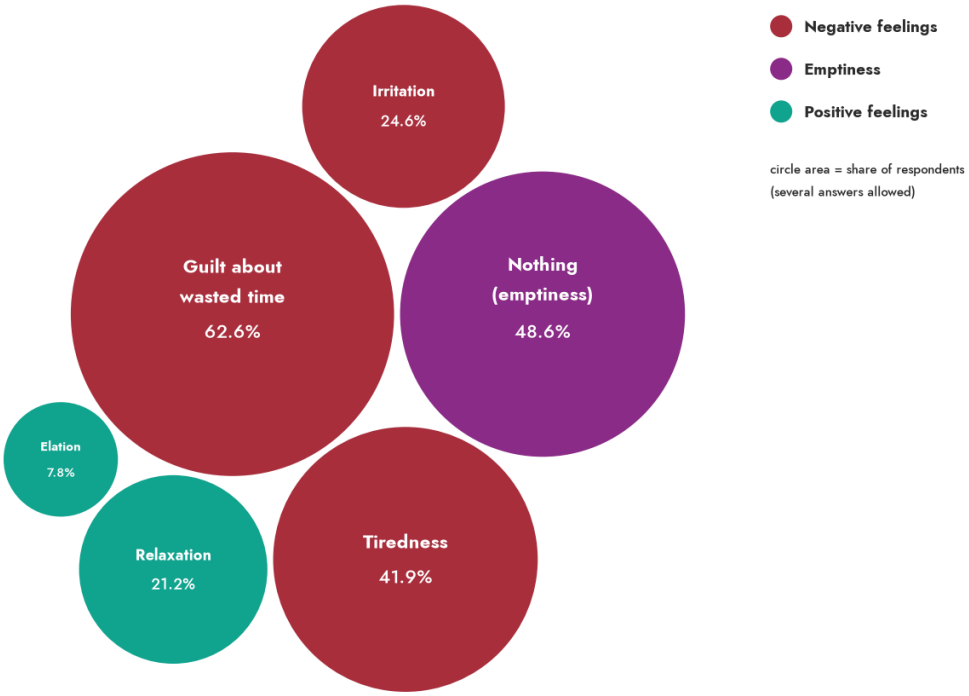
BRAIN FOG AFTER SCROLLING

How often do you feel "sensory overload" or brain fog after scrolling?



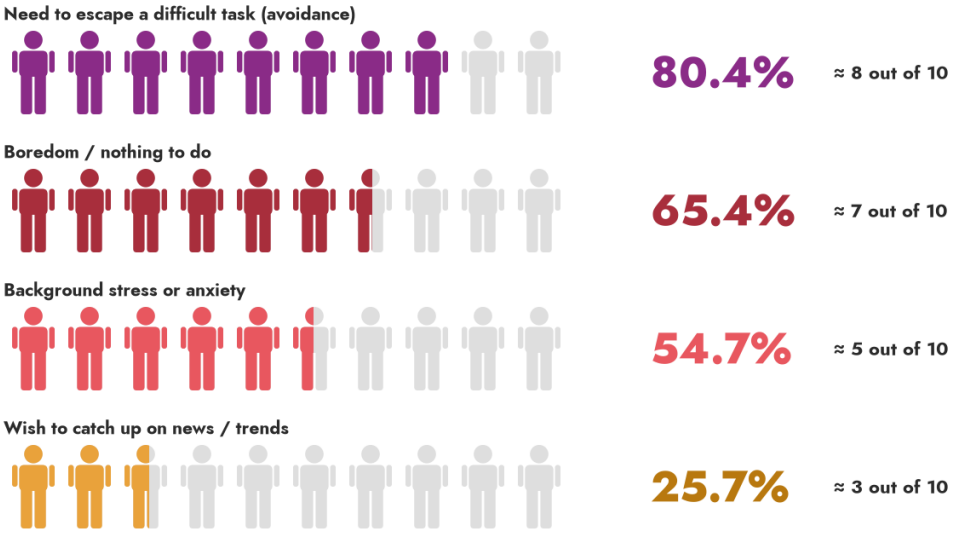
EMOTIONS AFTER SCROLLING

Which emotions do you feel most often right after long scrolling?



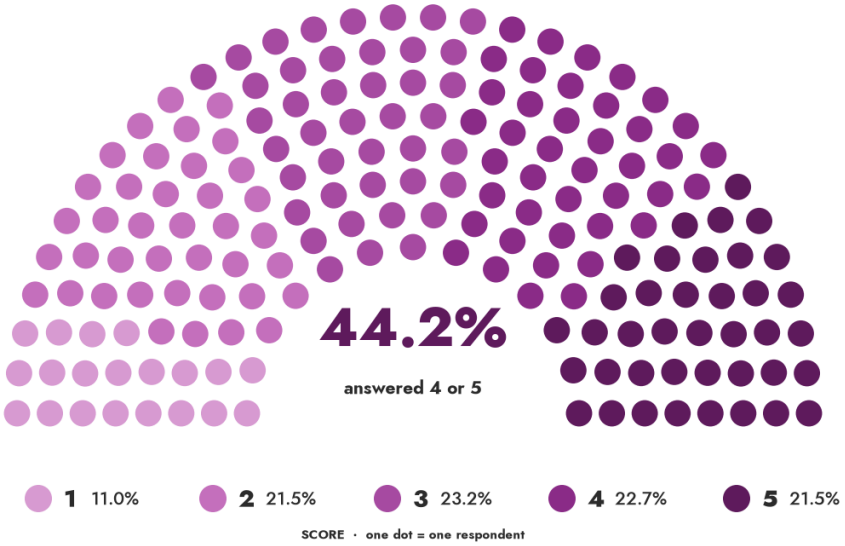
WHY PEOPLE OPEN THE FEED

What exactly makes you open the feed?



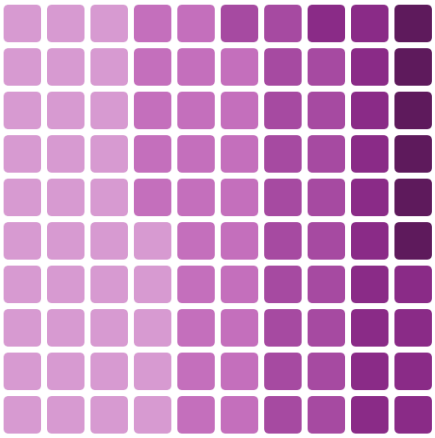
SCROLLING IN A CRISIS

Do you feel that your watch time increases during crisis situations?



COMMUNICATION SKILLS

Do you feel that your communication skills have got worse?



Score 1
35.4%

Score 2
23.8%

Score 3
20.4%

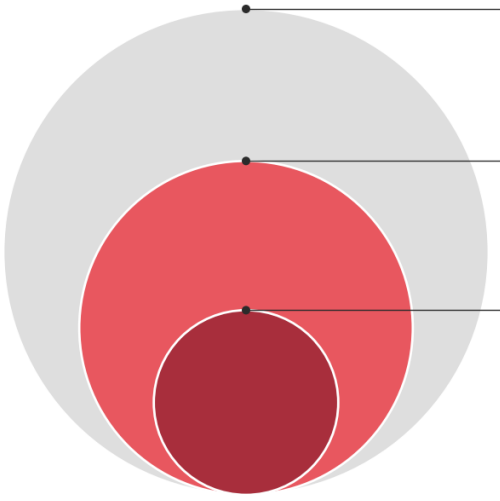
Score 4
14.9%

Score 5
5.5%

one square = 1% of respondents

IS IT THE SHORT VIDEOS?

Do you link this decline to watching short videos?



100% of respondents

47.2% feel their communication skills got worse

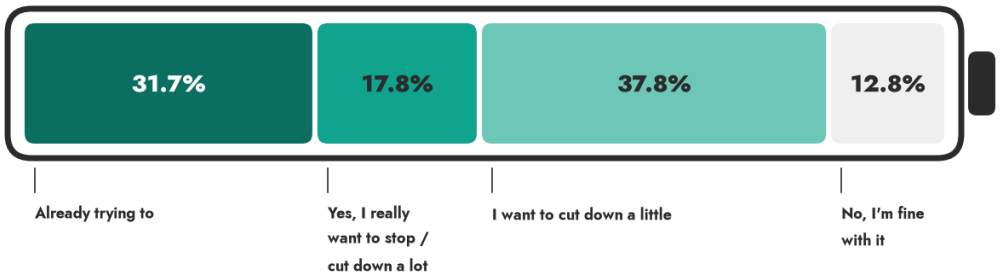
14.4% ...and blame short videos for it

that is 31% of those whose skills got worse

WISH TO SCROLL LESS

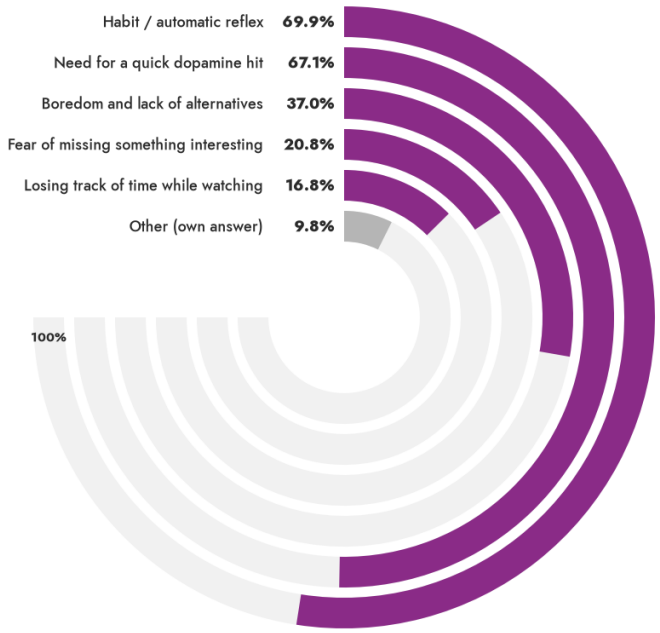
Do you want to reduce or completely stop scrolling short videos?

87.2% want to scroll less, or already try to



WHAT GETS IN THE WAY

If you want to watch less but can't, what gets in the way most?



5. Result

We have concluded exploratory data analysis of all our datasets. We have proven some or our hypothese right, and some of them wrong. Next step is to organize all visualization and information into logical and concise academic story.

